



Interfacial homogenization of a periodically corrugated surface in linear elasticity[☆]

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This paper is dedicated to Pedro Ponte Castañeda on the occasion of his 65th birthday

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ABSTRACT

This work presents a homogenization framework for modeling the mechanical behavior of three-dimensional linear elastic bodies with a periodically corrugated surface subjected to Dirichlet boundary conditions. The surface microstructure is assumed to be invariant along one spatial direction and periodic along the other. By combining asymptotic homogenization with matched asymptotic expansions near the surface corrugations, we derive an effective interface constitutive model that replaces the corrugated surface and the Dirichlet boundary condition with a flat boundary governed by a mixed (Robin-type) boundary condition. This boundary condition involves a second-order effective tensor, computed from elementary problems set on a representative periodic unit cell, hence allowing to account for the effect of the microstructure on the macroscopic response. We prove the symmetry and positive definiteness of the effective tensor and establish a uniqueness result of the effective problem. The model is assessed by comparison with 2D and 3D full-field simulations, demonstrating excellent agreement in both global and local responses. In particular, a cost-efficient post-processing strategy is proposed to reconstruct the local fields near the corrugations by use of a simple periodic unit cell, providing access to fine-scale information without the need for full-resolution computations.

1. Introduction

The present work addresses the interface homogenization of a semi-periodic corrugated boundary within the framework of linear elasticity, where the corrugation length scale is much smaller than the macroscopic dimension. Albeit small in amplitude, the corrugations significantly affect the overall deformation and stress fields in the structure. In order to capture accurately the fast displacement variations in the corrugated region, one typically has to resort to an extremely fine mesh leading to high computational cost. It is therefore of practical interest to develop a simplified, homogenized model that reproduces the full-field response – both in its global and local features – while substantially reducing computational effort.

Classical homogenization techniques (Sánchez-Palencia, 1980; Suquet, 1987; Ponte Castañeda and Zaidman, 1994; Ponte Castañeda and Willis, 1995) have been extensively developed for bulk microstructures—that is, heterogeneities distributed throughout the volume. These methods assume a bulk distribution of inclusions or heterogeneities, and thus are not directly applicable to problems where heterogeneities are localized on surfaces or interfaces. More recent work (Fergoug et al., 2022; Thbaut et al., 2024)

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extended these approaches by proposing higher-order asymptotic homogenization model for finite-sized composite materials that accounts for boundary layer effects in regions where boundary conditions are applied.

In contrast, interface homogenization, which deals with thin heterogeneous layers or interfaces whose thickness is much smaller than the macroscopic size, has been less extensively studied. Early rigorous work on boundary layer effects in elasticity is due to [Dumontet \(1986\)](#), who analyzed a linear elastic problem with Neumann boundary conditions. Several asymptotic models have been proposed to handle interface problems by exploiting a small parameter associated with the periodicity along the surface or interface ([Krasucki and Lenci, 2000](#); [Benveniste and Miloh, 2001](#); [Benveniste, 2006](#); [Bessoud et al., 2009](#)). At this point, one should also mention the use of half-space Green's function approaches to resolve or approximate analytically the problem of thin coatings and interfaces, such as the study of [Fleck and Willis \(2016\)](#) or more numerical conformal mapping approaches such as the one of [Vandembroucq and Roux \(1997\)](#). In turn, when the heterogeneity is periodic and localized at the boundary, its influence on displacement and stress fields appears typically at second order, necessitating asymptotic expansions up to that order to capture the effect. Matched asymptotic expansions have also been successfully applied to problems involving surface inhomogeneities or damage ([Nguetseng and Sánchez-Palencia, 1985](#); [Sánchez-Palencia and Suquet, 1983](#); [Abdelmoula and Marigo, 2000](#); [Sánchez-Palencia, 1987](#); [Marigo and Pideri, 2011](#); [David et al., 2012](#); [Fergoug et al., 2022](#)), and more recently to wave propagation in metamaterials ([Pham et al., 2017](#); [Marigo et al., 2017](#); [Pham et al., 2021](#)).

From a different perspective, [Eguiluz and Maradudin \(1983\)](#) proposed effective boundary conditions for wave propagation on rough surfaces using Green's function methods. Although analytically insightful, this approach is difficult to generalize, particularly for mixed boundary conditions and arbitrary corrugation shapes.

In most of these studies, the heterogeneous layer lies between two homogeneous media rather than at a boundary, and the specific case of corrugated boundary surfaces in static elasticity remains relatively unexplored. We propose here to establish the effective behavior of a corrugated surface of a linear elastic body subjected to Dirichlet boundary conditions by being attached to a rigid substrate. The proposed model includes first-order terms thus leading to much better estimates improving upon the zeroth order models presented earlier ([Sánchez-Palencia and Suquet, 1983](#); [Sánchez-Palencia, 1987](#)). The present model is capable of estimating mechanical fields even in case of problems with high gradients which is another novelty of the present model. Furthermore, an important feature of the proposed effective model is its capability to handle arbitrarily complex geometries of corrugations (e.g symmetric or asymmetric patterns) by using classical finite-element framework.

The manuscript is organized as follows. Section 2 presents the reference problem featuring a corrugated surface and its associated boundary conditions, as well as the corresponding effective problem with a Robin-type interface condition. Section 3 is devoted to the derivation of the effective model using a multiscale asymptotic analysis. In Section 4, we establish key properties of the homogenized tensor, and we prove the uniqueness of the solution to the effective problem based on its weak formulation. Finally, in Section 5, the effective model is validated through comparisons with direct numerical simulations across various test cases.

The notation used throughout the article is as follows: lowercase letters represent scalar variables, bold lowercase Latin letters denote vectors, bold lowercase Greek letters indicate second-order tensors, and bold uppercase Latin letters correspond to fourth-order tensors. In addition, we use $\bar{(\cdot)}$ to indicate quantities associated with the effective (homogenized macroscopic) problem, and $\tilde{(\cdot)}$ for the fields related to the microscopic periodic problem resolved at the level of a representative unit cell.

2. The full-field, effective and microscopic problem

In this section, we lay out the three boundary value problems treated in this work in the context of linear elasticity. Those involve the full-field, the effective and the microscopic problems. In the first one, depicted in [Fig. 1](#) (left domain), we resolve the full elasticity problem including the entire set of corrugations and the actual boundary conditions. This problem is used to validate the homogenization process. In turn, the effective (right domain in [Fig. 1](#)) and microscopic (periodic unit-cell in [Fig. 2](#)) problems are confronted together to provide the homogenized response of the corrugated interface.

We start by considering a three-dimensional body Ω made of an isotropic, linear elastic material, as shown in [Fig. 1](#). Denoting (σ, u) the corresponding Cauchy stress tensor and displacement vector, respectively, the governing equations in the full-field problem are (neglecting body forces and inertial terms)

$$\nabla \cdot \sigma = 0, \quad \sigma = C \varepsilon(u), \quad \varepsilon(u) = \frac{1}{2} (\nabla u + {}^T \nabla u) \quad \text{in } \Omega, \quad (1)$$

where C is the elasticity tensor, which is written as

$$C_{ijkl} = \left(\kappa - \frac{2\mu}{3} \right) \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \quad (2)$$

with $\kappa > 0$, $\mu > 0$ denoting the bulk modulus and shear modulus of the material, whereas δ_{ij} is the Kronecker delta. We assume that the lower surface of Ω located at $x_1 = 0$ is corrugated periodically with a period ℓ in the x_2 direction over a length L , while remains invariant along the x_3 direction over a length which is of the same order with L . These corrugations are bonded to a fixed rigid substrate that prescribes at $x_1 = 0$ a *constant* displacement U_D over a *non-empty* set Γ_D of the corrugated region. The remaining part of the boundary Γ_N of the corrugated region in contact with the air is traction free. These conditions read

$$\sigma n = 0 \quad \text{on } \Gamma_N, \quad u = U_D \quad \text{on } \Gamma_D \quad \text{with } U_D \text{ constant.} \quad (3)$$

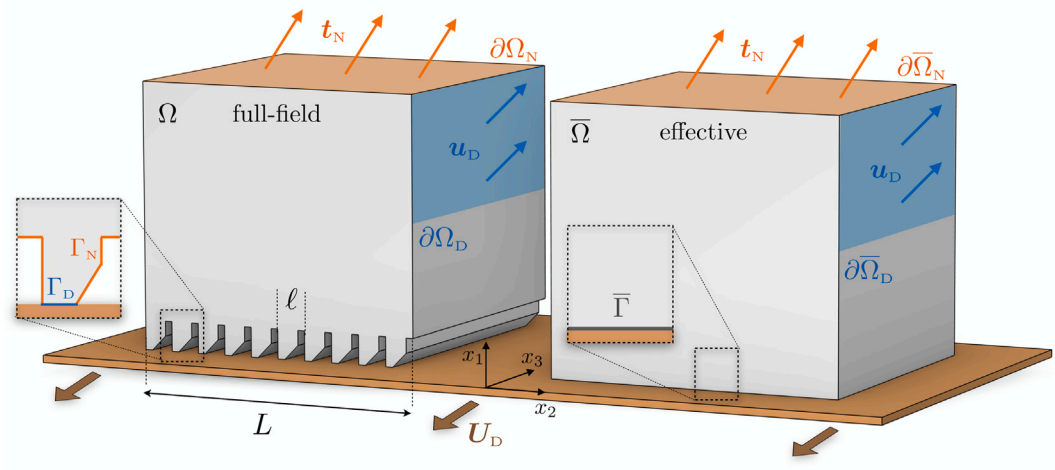


Fig. 1. Full-field problem (left) and effective problem (right) where the corrugated region attached to the rigid substrate at bottom has been replaced by the homogenized Robin-type boundary condition (5) applied on the flat region $\bar{\Gamma}$.

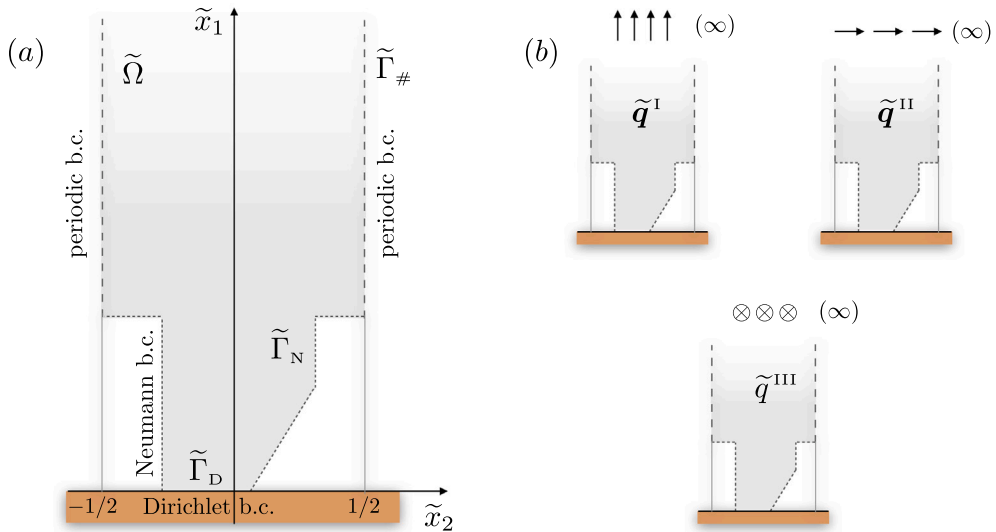


Fig. 2. (a) Elementary cell; (b) The three elementary problems $(\tilde{q}^I, \tilde{q}^{II}, \tilde{q}^{III})$.

Here, the subscripts “(N,D)” denote Neumann and Dirichlet parts, respectively. The full-field problem, as shown in Fig. 1 (left domain), is complemented with the following boundary conditions on the remaining boundaries far from the corrugated region

$$\sigma n = t_N \quad \text{on } \partial\Omega_N, \quad u = u_D \quad \text{on } \partial\Omega_D, \quad (4)$$

where the applied force t_N and u_D can vary in space.

Under the assumption of separation of length scales, implying here that $\ell \ll L$ and the prescribed displacement on Γ_D has spatial variation of the order of L , the above-defined boundary value problem (BVP) can be formally homogenized. We remark, however, for later reference, that even in cases that $\ell < L$ (and not much smaller), the model still provides accurate estimates, albeit with a gradually increasing error that becomes maximum when $\ell \sim L$.

Let $\bar{\Omega}$ denote the homogenized body, as shown in Fig. 1 (right domain), and $(\bar{\sigma}, \bar{u})$ the homogenized Cauchy stress tensor and displacement vector. Outside the corrugated region, bulk behavior (1) and relevant boundary conditions (4) remain identical and apply similarly on $(\bar{\sigma}, \bar{u})$ as in the full-field problem. Instead, the corrugated region is replaced by a corresponding Robin-type effective (or homogenized) boundary condition, set directly on a flat region $\bar{\Gamma}$ in contact with the rigid substrate

$$\bar{u}(0, \mathbf{x}') + \ell \bar{\beta} \bar{t}(0, \mathbf{x}') = U_D, \quad \mathbf{x}' = (x_2, x_3) \quad \text{on } \bar{\Gamma}, \quad (5)$$

where $\bar{\mathbf{t}}(0, \mathbf{x}') = \bar{\boldsymbol{\sigma}}(0, \mathbf{x}')\mathbf{n}$ is the normal traction at $x_1 = 0$ with $\mathbf{n} = -\mathbf{e}_1$ in the present problem and $\bar{\boldsymbol{\beta}}$ is a second order tensor that accounts for the effect of the microstructure and has the dimension of stress⁻¹. It takes the form

$$\bar{\boldsymbol{\beta}} = \bar{\mathbf{b}}^{\text{I}} \otimes \mathbf{e}_1 + \bar{\mathbf{b}}^{\text{II}} \otimes \mathbf{e}_2 + \bar{\mathbf{b}}^{\text{III}} \otimes \mathbf{e}_3 \quad (6)$$

where the vector coefficients $(\bar{\mathbf{b}}^{\text{I}}, \bar{\mathbf{b}}^{\text{II}}, \bar{\mathbf{b}}^{\text{III}})$ denoting effective parameters of the interface take the form

$$\bar{\mathbf{b}}^{\text{I}} = \bar{b}_1^{\text{I}} \mathbf{e}_1 + \bar{b}_2^{\text{I}} \mathbf{e}_2, \quad \bar{\mathbf{b}}^{\text{II}} = \bar{b}_1^{\text{II}} \mathbf{e}_1 + \bar{b}_2^{\text{II}} \mathbf{e}_2, \quad \bar{\mathbf{b}}^{\text{III}} = \bar{b}^{\text{III}} \mathbf{e}_3. \quad (7)$$

In the Proposition of Section 4.1, we prove that the effective tensor $\bar{\boldsymbol{\beta}}$ is symmetric and positive definite. This directly implies that the effective problem admits a variational formulation that is well-posed with unique solution thus allowing for a direct numerical implementation (as shown in Section 4). The coefficients defining $\bar{\boldsymbol{\beta}}$ are obtained from three independent elementary problems posed on a rescaled two-dimensional elementary cell with space variable $\tilde{\mathbf{x}} = \mathbf{x}/\ell$, which is periodic of unit length along $\mathbf{e}_2$ and semi-infinite along the $\mathbf{e}_1$ direction, as illustrated in Fig. 2.

These three independent elementary problems, defined in more detail in (30) and (31) in the next section, consist of:

- (I) an in-plane normal traction loading along $\mathbf{e}_1$ with *generalized* displacement solution $\tilde{\mathbf{q}}^{\text{I}}(\tilde{\mathbf{x}})$,
- (II) an in-plane shear loading along $\mathbf{e}_2$ with *generalized* displacement solution $\tilde{\mathbf{q}}^{\text{II}}(\tilde{\mathbf{x}})$, and,
- (III) an out-of-plane shear loading along $\mathbf{e}_3$ with *generalized* displacement solution $\tilde{\mathbf{q}}^{\text{III}}(\tilde{\mathbf{x}})$.

We note here that the generalized displacements $(\tilde{\mathbf{q}}^{\text{I}}, \tilde{\mathbf{q}}^{\text{II}}, \tilde{\mathbf{q}}^{\text{III}})$ have an inverse of stress dimension (i.e. m²/N). Then, after solving these generalized displacement fields (see Appendix A for the numerical computation procedure), the effective coefficients (7) are estimated by taking the limit $\tilde{x}_1 \rightarrow +\infty$, such that

$$\bar{\mathbf{b}}^{\text{I}} = \lim_{\tilde{x}_1 \rightarrow +\infty} \left(\tilde{\mathbf{q}}^{\text{I}} - \frac{\tilde{x}_1}{\kappa + \frac{4\mu}{3}} \mathbf{e}_1 \right), \quad \bar{\mathbf{b}}^{\text{II}} = \lim_{\tilde{x}_1 \rightarrow +\infty} \left(\tilde{\mathbf{q}}^{\text{II}} - \frac{\tilde{x}_1}{\mu} \mathbf{e}_2 \right), \quad \bar{\mathbf{b}}^{\text{III}} = \lim_{\tilde{x}_1 \rightarrow +\infty} \left(\tilde{\mathbf{q}}^{\text{III}} - \frac{\tilde{x}_1}{\mu} \right). \quad (8)$$

This procedure allows us to interpret these quantities as boundary layer coefficients capturing the influence of the surface microstructure. Note that in the case of symmetric corrugations with respect to the axis x_2 , the effective tensor $\bar{\boldsymbol{\beta}}$ becomes diagonal as we have $\bar{b}_1^{\text{II}} = \bar{b}_2^{\text{I}} = 0$, see Proposition in Section 4. In the next section, we provide a detailed derivation of these results.

3. Asymptotic derivation of the effective problem

3.1. Setting of the asymptotic procedure

We adopt a two-scale homogenization analysis coupled with matched asymptotic expansions to derive the effective boundary conditions that will replace the corrugated region attached to the rigid substrate in the limit problem. Given the characteristic macroscopic length L of the corrugated region (see Fig. 1), we introduce henceforth the rescaled variables

$$\mathbf{x} \mapsto \frac{\mathbf{x}}{L}, \quad \mathbf{u} \mapsto \frac{\mathbf{u}}{L}, \quad (9)$$

as well as the rescaled periodicity η given by

$$\eta = \frac{\ell}{L} \ll 1. \quad (10)$$

The parameter η is the small parameter in the BVP and thus in the asymptotic analysis, discussed in the following. In the limit of small η , a separation of length scales occurs and the direct problem can be split into two different regions: the macroscopic region and the microscopic region.

3.1.1. Macroscopic region

At the macroscopic scale, the corrugated region *collapses* to a *flat* boundary located at the plane $x_1 = 0$. We then consider a classical expansion of the solution fields $(\mathbf{u}, \boldsymbol{\sigma})$ at this scale as

$$\mathbf{u} = \sum_{n \geq 0} \eta^n \bar{\mathbf{u}}^n(\mathbf{x}), \quad \boldsymbol{\sigma} = \sum_{n \geq 0} \eta^n \bar{\boldsymbol{\sigma}}^n(\mathbf{x}) \quad \text{in } \bar{\Omega}, \quad (11)$$

as well as of the applied boundary terms,

$$\mathbf{t}_N = \sum_{n \geq 0} \eta^n \bar{\mathbf{t}}_N^n \quad \text{on } \partial\bar{\Omega}_N, \quad \mathbf{u}_D = \sum_{n \geq 0} \eta^n \bar{\mathbf{u}}_D^n \quad \text{on } \partial\bar{\Omega}_D. \quad (12)$$

Injecting expansions (11) in the equilibrium equation, constitutive relations and compatibility conditions given in (1), we get for $n \geq 0$,

$$\nabla \cdot \bar{\boldsymbol{\sigma}}^n = \mathbf{0}, \quad \bar{\boldsymbol{\sigma}}^n = C \boldsymbol{\varepsilon}(\bar{\mathbf{u}}^n), \quad \boldsymbol{\varepsilon}(\bar{\mathbf{u}}^n) = \frac{1}{2} (\nabla(\bar{\mathbf{u}}^n) + {}^T \nabla(\bar{\mathbf{u}}^n)), \quad \text{in } \bar{\Omega}, \quad (13)$$

where $(\nabla, \boldsymbol{\varepsilon})$ refer to differential operators calculated with respect to the macroscopic variable $\mathbf{x}$. Next, injecting (11) in the boundary conditions (4), we find for $n \geq 0$,

$$\bar{\boldsymbol{\sigma}}^n \mathbf{n} = \bar{\mathbf{t}}_N^n \quad \text{on } \partial\bar{\Omega}_N, \quad \bar{\mathbf{u}}^n = \bar{\mathbf{u}}_D^n \quad \text{on } \partial\bar{\Omega}_D. \quad (14)$$

At this macroscopic scale described by the coordinate system $\mathbf{x}$, the remaining unknowns are those describing the boundary conditions that need to be applied at $x_1 = 0$. They will be obtained from the matched asymptotic analysis between the macro- and the micro-regions discussed below in Section 3.2.

3.1.2. Microscopic region

Considering a zoomed region of size $1/\eta$ at a point $\mathbf{x} = (0, \mathbf{x}')$ with $\mathbf{x}' = (x_2, x_3)$ on the effective boundary $\bar{\Gamma}$, we reveal a periodic array of corrugations subjected to a *constant* prescribed displacement U_D . Taking into account the invariance of the corrugations along the x_3 -direction, one may describe this microscopic (zoomed) region with the help of a rescaled coordinate $\tilde{\mathbf{x}} = (\tilde{x}_1, \tilde{x}_2) = (x_1/\eta, x_2/\eta)$ that varies in a two-dimensional semi-infinite elementary cell $\tilde{\Omega}$ depicted in Fig. 2 and defined as

$$\tilde{\Omega} = \mathbb{R}^+ \times \left(-\frac{1}{2}, \frac{1}{2}\right) \setminus S, \quad (15)$$

where S corresponds to the void region surrounding the corrugation within the elementary cell $\tilde{\Omega}$. The solution fields $(\mathbf{u}, \boldsymbol{\sigma})$ may then be expanded in this microscopic region as

$$\mathbf{u} = \sum_{n \geq 0} \eta^n \tilde{\mathbf{u}}^n(\tilde{\mathbf{x}}, \mathbf{x}'), \quad \boldsymbol{\sigma} = \sum_{n \geq 0} \eta^n \tilde{\boldsymbol{\sigma}}^n(\tilde{\mathbf{x}}, \mathbf{x}') \quad \text{in } \tilde{\Omega}, \quad (16)$$

with $(\tilde{\mathbf{u}}^n, \tilde{\boldsymbol{\sigma}}^n)$ being $\tilde{x}_2$ -periodic functions of period 1 on $\tilde{\Gamma}_\#$. At the bottom of the periodic cell and in accordance with the boundary conditions applied in the full-field problem defined in (4), the corrugation in the microscopic problem is also subjected to a constant prescribed displacement U_D over a non-empty rescaled boundary $\tilde{\Gamma}_D$ such that

$$U_D = \sum_{n \geq 0} \eta^n U_D^n \quad \text{on } \tilde{\Gamma}_D, \quad (17)$$

while the remaining boundary $\tilde{\Gamma}_N$ is traction free. From the expansion (16), we readily deduce that

$$n \geq 0, \quad \tilde{\mathbf{u}}^n = U_D^n \quad \text{on } \tilde{\Gamma}_D, \quad \tilde{\boldsymbol{\sigma}}^n \mathbf{n} = \mathbf{0} \quad \text{on } \tilde{\Gamma}_N. \quad (18)$$

At the microscopic level, the gradient operator is given by $\nabla \mapsto \eta^{-1} \tilde{\nabla} + \nabla'$ with a contribution $\eta^{-1} \tilde{\nabla} = \eta^{-1} \partial(\cdot)/\partial \tilde{\mathbf{x}}$ with respect to the rescaled coordinate $\tilde{\mathbf{x}}$ (the factor η^{-1} coming from a simple chain rule since $(\tilde{x}_1, \tilde{x}_2) = (x_1/\eta, x_2/\eta)$) and a contribution $\nabla' = \partial(\cdot)/\partial \mathbf{x}'$ with respect to the macroscopic in-plane coordinate $\mathbf{x}'$. Similarly the strain tensor in the microscopic region is given by as $\boldsymbol{\varepsilon} \mapsto \eta^{-1} \tilde{\boldsymbol{\varepsilon}} + \boldsymbol{\varepsilon}'$.

The hierarchy of governing equations are obtained by formally substituting the expanded solution (16) and the rescaled space operators for the microscopic region in (1) and then isolating each term of the expansion order-by-order to obtain at the order η^{-1} ,

$$\mathbf{0} = C \tilde{\boldsymbol{\varepsilon}}(\tilde{\mathbf{u}}^0), \quad \tilde{\nabla} \cdot \tilde{\boldsymbol{\sigma}}^0 = \mathbf{0} \quad \text{in } \tilde{\Omega}, \quad (19)$$

and at the order η^0 ,

$$\tilde{\boldsymbol{\sigma}}^0 = C \left(\tilde{\boldsymbol{\varepsilon}}(\tilde{\mathbf{u}}^1) + \boldsymbol{\varepsilon}'(\tilde{\mathbf{u}}^0) \right), \quad \tilde{\nabla} \cdot \tilde{\boldsymbol{\sigma}}^1 + \nabla' \cdot \tilde{\boldsymbol{\sigma}}^0 = \mathbf{0} \quad \text{in } \tilde{\Omega}. \quad (20)$$

Remark. The present asymptotic analysis is restricted to regions where the surface microstructure remains locally periodic. The micro-to-macro behavior near the sample corners, where this assumption breaks down, lies beyond the scope of the current work. A dedicated asymptotic treatment would be required to quantify their contribution to the effective properties. Nevertheless, as demonstrated in Section 5, neglecting these corner effects does not compromise the overall accuracy of the proposed homogenized model. In fact, the model still provides fairly acceptable estimates near these boundaries for any practical use.

3.2. Matching conditions

The macroscopic and microscopic displacement and stress fields introduced above must be asymptotically matched in an intermediate overlapping region, corresponding to the limit $x_1 \rightarrow 0$ in the macroscopic coordinates and $\tilde{x}_1 \rightarrow +\infty$ in the microscopic (rescaled) coordinates, i.e.,

$$\tilde{\mathbf{u}}^0(\mathbf{x}) + \eta \tilde{\mathbf{u}}^1(\mathbf{x}) + \dots \underset{x_1 \rightarrow 0, \tilde{x}_1 \rightarrow +\infty}{\sim} \tilde{\mathbf{u}}^0(\tilde{\mathbf{x}}, \mathbf{x}') + \eta \tilde{\mathbf{u}}^1(\tilde{\mathbf{x}}, \mathbf{x}') + \dots \quad (21)$$

Using Taylor series expansion for $\mathbf{u}^n$ ($n \geq 0$) about $x_1 = 0$, we get

$$\tilde{\mathbf{u}}^0(\eta \tilde{x}_1, \mathbf{x}') + \eta \tilde{\mathbf{u}}^1(\eta \tilde{x}_1, \mathbf{x}') + \dots \sim \tilde{\mathbf{u}}^0(0, \mathbf{x}') + \eta \left[\tilde{\mathbf{u}}^1(0, \mathbf{x}') + \tilde{x}_1 \frac{\partial \tilde{\mathbf{u}}^0}{\partial \tilde{x}_1}(0, \mathbf{x}') \right] + \dots \quad (22)$$

The matching conditions between the macro- and microscopic displacement fields are obtained by enforcing equality of the right hand sides of (21) and (22) for the terms with the same order η , such that

$$\begin{cases} \tilde{\mathbf{u}}^0(0, \mathbf{x}') = \lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\mathbf{u}}^0(\tilde{\mathbf{x}}, \mathbf{x}'), \\ \tilde{\mathbf{u}}^1(0, \mathbf{x}') = \lim_{\tilde{x}_1 \rightarrow +\infty} \left(\tilde{\mathbf{u}}^1(\tilde{\mathbf{x}}, \mathbf{x}') - \tilde{x}_1 \frac{\partial \tilde{\mathbf{u}}^0}{\partial \tilde{x}_1}(0, \mathbf{x}') \right). \end{cases} \quad (23)$$

Similarly, the matching conditions for micro- and macro-stress fields can be obtained as

$$\begin{cases} \bar{\sigma}^0(0, \mathbf{x}') = \lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\sigma}^0(\tilde{\mathbf{x}}, \mathbf{x}'), \\ \bar{\sigma}^1(0, \mathbf{x}') = \lim_{\tilde{x}_1 \rightarrow +\infty} \left(\tilde{\sigma}^1(\tilde{\mathbf{x}}, \mathbf{x}') - \tilde{x}_1 \frac{\partial \tilde{\sigma}^0}{\partial \tilde{x}_1}(0, \mathbf{x}') \right). \end{cases} \quad (24)$$

These set of conditions (23) and (24) will be used subsequently to solve for the 1st-order fields in terms of the 0th-order fields. Note that the approximate symbol is used to indicate that only a small number of terms will be retained in the expansion, whereas one may keep more terms if the problem requires such an operation. Note, however, that the more terms we keep the more complex will be the final model, without necessarily adding a lot to the accuracy of the prediction.

3.2.1. Dominant 0_{th}-order : Dirichlet boundary condition

From the first equation of (19), we deduce that $\tilde{\epsilon}(\tilde{\mathbf{u}}^0) = 0$ as C is invertible, hence the microscopic displacement $\tilde{\mathbf{u}}^0$ is a rigid body motion with respect to the microscopic variable $\tilde{\mathbf{x}}$ and can be written as

$$\tilde{\mathbf{u}}^0(\tilde{\mathbf{x}}, \mathbf{x}') = \mathbf{a}(\mathbf{x}') + \omega(\mathbf{x}') \mathbf{e}_3 \wedge \tilde{\mathbf{x}} \quad \text{in } \tilde{\Omega}. \quad (25)$$

Given the periodicity condition on $\tilde{\Gamma}_\#$ satisfied by $\tilde{\mathbf{u}}^0$, we deduce that $\omega(\mathbf{x}') = 0$. Next given the Dirichlet boundary condition on $\tilde{\Gamma}_D$, see (18), we find that $\mathbf{a}(\mathbf{x}') = \mathbf{U}_D^0$ is a constant equal to the prescribed displacement. It remains to apply the matching condition at order 0 in (23) to obtain at the leading-order a classical Dirichlet boundary condition

$$\bar{\mathbf{u}}^0(0, \mathbf{x}') = \mathbf{U}_D^0 \quad \text{on } \bar{\Gamma}, \quad (26)$$

which means that, at the zeroth order, the entire effective surface is subjected to a uniform Dirichlet condition—as if the body were in full contact with the flat support and the surface corrugations were completely absent.

3.2.2. First order correction: Robin-type boundary condition

At 1st-order, the displacement $\tilde{\mathbf{u}}^1$ and the stress $\tilde{\sigma}^0$ need to satisfy at the microscopic level the following BVP problem

$$\begin{cases} \tilde{\nabla} \cdot \tilde{\sigma}^0 = \mathbf{0}, & \tilde{\sigma}^0 = C\tilde{\epsilon}(\tilde{\mathbf{u}}^1) \quad \text{in } \tilde{\Omega} \\ \tilde{\mathbf{u}}^1 = \mathbf{U}_D^1 \quad \text{on } \tilde{\Gamma}_D, & \tilde{\sigma}^0 \mathbf{n} = \mathbf{0} \quad \text{on } \tilde{\Gamma}_N, \\ \lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\sigma}^0(\tilde{\mathbf{x}}, \mathbf{x}') \mathbf{n} = \bar{\sigma}^0(0, \mathbf{x}') \mathbf{n}, \\ \tilde{\mathbf{u}}^1 : \tilde{\Omega}_2 - \text{periodic}, & \tilde{\sigma}^0 \mathbf{n} : \tilde{\Omega}_2 - \text{anti-periodic} \quad \text{on } \tilde{\Gamma}_\#. \end{cases} \quad (27)$$

Here, we have used the previous 0th-order solution, i.e., that $\tilde{\epsilon}(\tilde{\mathbf{u}}^0) = \mathbf{0}$ since $\tilde{\mathbf{u}}^0 = \mathbf{U}_D^0$. Solving the microscopic BVP problem (27) consists in determining the explicit dependency of $(\tilde{\mathbf{u}}^1, \tilde{\sigma}^0)$ on $\tilde{\mathbf{x}}$. We distinguish two types of macroscopic quantities acting as an external loading in the microscopic problem (27):

- the prescribed displacement $\mathbf{U}_D^1$ applied on $\tilde{\Gamma}_D$, and
- the macroscopic stress $\bar{\sigma}^0(0, \mathbf{x}')$ applied at infinity when $\tilde{x}_1 \rightarrow +\infty$. Since $\frac{\partial \tilde{\mathbf{u}}^0}{\partial \tilde{x}'}(0, \mathbf{x}') = \mathbf{0}$ from (26), we deduce from (13) for $n = 0$ that

$$\bar{\sigma}_{22}^0(0, \mathbf{x}') = \bar{\sigma}_{33}^0(0, \mathbf{x}') = \frac{3\kappa - 2\mu}{3\kappa + 4\mu} \bar{\sigma}_{11}^0(0, \mathbf{x}'), \quad \bar{\sigma}_{23}^0(0, \mathbf{x}') = 0 \quad \text{on } \bar{\Gamma}. \quad (28)$$

This leaves us with three remaining independent macroscopic stress components $\bar{\sigma}_{11}^0(0, \mathbf{x}')$, $\bar{\sigma}_{12}^0(0, \mathbf{x}')$ and $\bar{\sigma}_{13}^0(0, \mathbf{x}')$ in the problem (27).

Note that given a set of macroscopic quantities $(\mathbf{U}_D^1, \bar{\sigma}_{li}^0(0, \mathbf{x}'))$, the solution of (27) with respect to $\tilde{\mathbf{x}}$ is uniquely defined since a Dirichlet BC is prescribed on $\tilde{\Gamma}_D$. For the sake of clarity, we here anticipate (and this can be verified by direct substitution) that the solution $\tilde{\mathbf{u}}^1$ of (27) can be expanded by linearity with respect to $(\mathbf{U}_D^1, \bar{\sigma}_{li}^0(0, \mathbf{x}'))$ as

$$\tilde{\mathbf{u}}^1(\tilde{\mathbf{x}}, \mathbf{x}') = \mathbf{U}_D^1 + \bar{\sigma}_{11}^0(0, \mathbf{x}') \tilde{\mathbf{q}}^I(\tilde{\mathbf{x}}) + \bar{\sigma}_{12}^0(0, \mathbf{x}') \tilde{\mathbf{q}}^{II}(\tilde{\mathbf{x}}) + \bar{\sigma}_{13}^0(0, \mathbf{x}') \tilde{\mathbf{q}}^{III}(\tilde{\mathbf{x}}) \mathbf{e}_3 \quad \text{in } \tilde{\Omega}. \quad (29)$$

In (29), the fields $(\tilde{\mathbf{q}}^I(\tilde{\mathbf{x}}), \tilde{\mathbf{q}}^{II}(\tilde{\mathbf{x}}))$ are two-dimensional *generalized* displacements (with $\tilde{\mathbf{q}}_3^I = \tilde{\mathbf{q}}_3^{II} = 0$ and dimension stress⁻¹) solving the unitary in-plane problems

$$i = I, II \quad \begin{cases} \tilde{\nabla}^{2D} \cdot \tilde{\boldsymbol{\tau}}^i = \mathbf{0}, & \tilde{\boldsymbol{\tau}}^i = C\tilde{\epsilon}^{2D}(\tilde{\mathbf{q}}^i) \quad \text{in } \tilde{\Omega}, \\ \tilde{\mathbf{q}}^i : \tilde{\Omega}_2 - \text{periodic}, & \tilde{\boldsymbol{\tau}}^i \mathbf{n} : \tilde{\Omega}_2 - \text{anti-periodic} \quad \text{on } \tilde{\Gamma}_\#, \\ \tilde{\mathbf{q}}^i = \mathbf{0} \quad \text{on } \tilde{\Gamma}_D, & \tilde{\boldsymbol{\tau}}^i \mathbf{n} = \mathbf{0} \quad \text{on } \tilde{\Gamma}_N, \\ \lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\boldsymbol{\tau}}^i \mathbf{e}_1 = \mathbf{e}_1, & \lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\boldsymbol{\tau}}^{II} \mathbf{e}_1 = \mathbf{e}_2. \end{cases} \quad (30)$$

In these expressions, $(\tilde{\boldsymbol{\tau}}^I(\tilde{\mathbf{x}}), \tilde{\boldsymbol{\tau}}^{II}(\tilde{\mathbf{x}}))$ are the associated *generalized* stress fields (which are non-dimensional) associated with the generalized displacements $\tilde{\mathbf{q}}^i$, whereby $\tilde{\epsilon}^{2D}(\cdot)$ is the two-dimensional, in-plane *generalized* strain operator (with dimension stress⁻¹).

Next in (29), $\tilde{q}^{\text{III}}(\tilde{\mathbf{x}})$ is a scalar field solving the out-of-plane problem,

$$\begin{cases} \tilde{\nabla}^{2\text{D}} \cdot \tilde{\boldsymbol{\tau}}^{\text{III}} = 0, & \tilde{\boldsymbol{\tau}}^{\text{III}} = \mu \tilde{\nabla}^{2\text{D}} \tilde{q}^{\text{III}} \quad \text{in } \tilde{\Omega}, \\ \tilde{q}^{\text{III}} : \tilde{\mathbf{x}}_2 - \text{periodic}, & \tilde{\boldsymbol{\tau}}^{\text{III}} \cdot \mathbf{n} : \tilde{\mathbf{x}}_2 - \text{anti-periodic} \quad \text{on } \tilde{\Gamma}_\#, \\ \tilde{q}^{\text{III}} = 0 \quad \text{on } \tilde{\Gamma}_\text{D}, & \tilde{\boldsymbol{\tau}}^{\text{III}} \cdot \mathbf{n} = 0 \quad \text{on } \tilde{\Gamma}_\text{N}, \\ \lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\boldsymbol{\tau}}^{\text{III}} \cdot \mathbf{e}_1 = 1. \end{cases} \quad (31)$$

with $\tilde{\boldsymbol{\tau}}^{\text{III}}(\tilde{\mathbf{x}})$ being the associated generalized stress field and $\tilde{\nabla}^{2\text{D}}(\cdot)$ being the two-dimensional generalized gradient operator.

For the final determination of the effective boundary condition of the interface, one needs to know the behavior of $(\tilde{q}^{\text{I}}, \tilde{q}^{\text{II}}, \tilde{q}^{\text{III}})$ at infinity. This allows then to apply the matching condition (23) on the displacement fields at order 1. Inverting the constitutive behavior in (30) and (31) and making use of the behavior at infinity of $\tilde{\boldsymbol{\tau}}^i$, we obtain for $i = \text{I, II}$,

$$\lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\boldsymbol{\varepsilon}}^{2\text{D}}(\tilde{q}^{\text{I}}) = \frac{1}{\kappa + \frac{4\mu}{3}} \mathbf{e}_1 \otimes \mathbf{e}_1, \quad \lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\boldsymbol{\varepsilon}}^{2\text{D}}(\tilde{q}^{\text{II}}) = \frac{1}{2\mu} (\mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1) \quad (32)$$

and for $i = \text{III}$,

$$\lim_{\tilde{x}_1 \rightarrow +\infty} \tilde{\nabla}^{2\text{D}} \tilde{q}^{\text{III}} = \frac{1}{\mu} \mathbf{e}_1. \quad (33)$$

This enforces the behavior of $(\tilde{q}^{\text{I}}, \tilde{q}^{\text{II}}, \tilde{q}^{\text{III}})$ at infinity to be of the form

$$\tilde{q}^{\text{I}} \underset{\tilde{x}_1 \rightarrow +\infty}{\sim} \frac{\tilde{x}_1}{\kappa + \frac{4\mu}{3}} \mathbf{e}_1 + \bar{\mathbf{b}}^{\text{I}}, \quad \tilde{q}^{\text{II}} \underset{\tilde{x}_1 \rightarrow +\infty}{\sim} \frac{\tilde{x}_1}{\mu} \mathbf{e}_2 + \bar{\mathbf{b}}^{\text{II}}, \quad \tilde{q}^{\text{III}} \underset{\tilde{x}_1 \rightarrow +\infty}{\sim} \frac{\tilde{x}_1}{\mu} + \bar{\mathbf{b}}^{\text{III}}. \quad (34)$$

In this expression, $(\bar{\mathbf{b}}^{\text{I}}, \bar{\mathbf{b}}^{\text{II}}, \bar{\mathbf{b}}^{\text{III}})$ denote boundary layer constants and can be directly interpreted as effective coefficients of the boundary surface accounting for the effect of the corrugation. It now suffices to combine (29) and (34) to find the behavior at infinity of $\tilde{\mathbf{u}}^{\text{I}}$, such that

$$\begin{aligned} \tilde{\mathbf{u}}^{\text{I}}(\tilde{\mathbf{x}}, \mathbf{x}') \underset{\tilde{x}_1 \rightarrow +\infty}{\sim} & \mathbf{U}_\text{D}^1 + \bar{\sigma}_{11}^0(0, \mathbf{x}') \bar{\mathbf{b}}^{\text{I}} + \bar{\sigma}_{12}^0(0, \mathbf{x}') \bar{\mathbf{b}}^{\text{II}} + \bar{\sigma}_{13}^0(0, \mathbf{x}') \bar{\mathbf{b}}^{\text{III}} \mathbf{e}_3 \\ & + \left(\frac{\bar{\sigma}_{11}^0(0, \mathbf{x}')}{\kappa + \frac{4\mu}{3}} \mathbf{e}_1 + \frac{\bar{\sigma}_{12}^0(0, \mathbf{x}')}{\mu} \mathbf{e}_2 + \frac{\bar{\sigma}_{13}^0(0, \mathbf{x}')}{\mu} \mathbf{e}_3 \right) \tilde{x}_1. \end{aligned} \quad (35)$$

Next, given the constitutive behavior (13) for $n = 0$ and the boundary condition (26), we get

$$\bar{\sigma}_{11}^0(0, \mathbf{x}') = \left(\kappa + \frac{4\mu}{3} \right) \frac{\partial \bar{u}_1^0}{\partial x_1}(0, \mathbf{x}'), \quad \bar{\sigma}_{1i}^0(0, \mathbf{x}') = \mu \frac{\partial \bar{u}_i^0}{\partial x_1}(0, \mathbf{x}') \quad i = 2, 3 \quad \text{on } \bar{\Gamma}. \quad (36)$$

Eventually, combining (35)–(36) and applying the matching condition (23) at order 1 leads to

$$\bar{\mathbf{u}}^1(0, \mathbf{x}') = \mathbf{U}_\text{D}^1 - \bar{\boldsymbol{\beta}} \bar{\mathbf{t}}^0(0, \mathbf{x}') \quad \text{on } \bar{\Gamma}, \quad (37)$$

where $\bar{\mathbf{t}}^0(0, \mathbf{x}') = \bar{\boldsymbol{\sigma}}^0(0, \mathbf{x}')(-\mathbf{e}_1)$ is the normal traction and $\bar{\boldsymbol{\beta}}$ is the homogenized second-order tensor $\bar{\boldsymbol{\beta}}$ introduced in (6).

3.3. Unique formulation and effective boundary condition

Owing to the linear elastic framework adopted in the present work, the final macroscopic problem is written in terms of the unique fields $(\bar{\mathbf{u}}, \bar{\boldsymbol{\sigma}})$

$$\bar{\mathbf{u}} = \bar{\mathbf{u}}^0 + \eta \bar{\mathbf{u}}^1 + O(\eta^2), \quad \bar{\boldsymbol{\sigma}} = \bar{\boldsymbol{\sigma}}^0 + \eta \bar{\boldsymbol{\sigma}}^1 + O(\eta^2) \quad \text{in } \bar{\Omega}, \quad (38)$$

with the homogenized body $\bar{\Omega}$ submitted to the set of loadings

$$\mathbf{u}_\text{D} = \bar{\mathbf{u}}_\text{D}^0 + \eta \bar{\mathbf{u}}_\text{D}^1 + O(\eta^2) \quad \text{on } \partial \bar{\Omega}_\text{D}, \quad \mathbf{t}_\text{N} = \bar{\mathbf{t}}_\text{N}^0 + \eta \bar{\mathbf{t}}_\text{N}^1 + O(\eta^2) \quad \text{on } \partial \bar{\Omega}_\text{N}. \quad (39)$$

Using (13), the bulk behavior of $(\bar{\mathbf{u}}, \bar{\boldsymbol{\sigma}})$ is governed by the BVP, neglecting terms of order $O(\eta^2)$,

$$\begin{cases} \nabla \cdot \bar{\boldsymbol{\sigma}} = \mathbf{0}, & \bar{\boldsymbol{\sigma}} = C \boldsymbol{\varepsilon}(\bar{\mathbf{u}}), \quad \boldsymbol{\varepsilon}(\bar{\mathbf{u}}) = \frac{1}{2} (\nabla(\bar{\mathbf{u}}) + {}^T \nabla(\bar{\mathbf{u}})) \quad \text{in } \bar{\Omega}, \\ \bar{\boldsymbol{\sigma}} \mathbf{n} = \mathbf{t}_\text{N} \quad \text{on } \partial \bar{\Omega}_\text{N}, & \bar{\mathbf{u}} = \mathbf{u}_\text{D} \quad \text{on } \partial \bar{\Omega}_\text{D}. \end{cases} \quad (40)$$

At the interface with the rigid substrate at $x_1 = 0$, the effective boundary condition for the unique formulation is obtained by adding the contributions (26) at order 0 and (37) at order 1 as well as neglecting terms of order $O(\eta^2)$,

$$\bar{\mathbf{u}}(0, \mathbf{x}') + \eta \bar{\boldsymbol{\beta}} \bar{\mathbf{t}}(0, \mathbf{x}') = \mathbf{U}_\text{D} \quad \text{on } \bar{\Gamma}. \quad (41)$$

The above equation is the normalized version of (5). To recover its dimensional expression, we use (9) and (10), so that (41) can be rewritten as

$$\frac{\bar{\mathbf{u}}(0, \mathbf{x}')}{L} + \frac{\ell}{L} \bar{\boldsymbol{\beta}} \bar{\mathbf{t}}(0, \mathbf{x}') = \frac{\mathbf{U}_\text{D}}{L} \quad \text{on } \bar{\Gamma}. \quad (42)$$

Multiplying the above expression by L then yields the expected dimensional form of the effective boundary condition announced in (5), i.e.,

$$\bar{u}(0, \mathbf{x}') + \ell \bar{\beta} \bar{t}(0, \mathbf{x}') = U_D \quad \text{on } \bar{\Gamma}.$$

4. Properties on the effective problem

4.1. Symmetry and definite positivity of the homogenized tensor $\bar{\beta}$

In this section, we establish some useful properties on the homogenized tensor $\bar{\beta}$:

Proposition. The homogenized tensor $\bar{\beta}$ is symmetric and definite positive. In particular, we have the following variational inequality,

$$\forall \mathbf{a} \in \mathbb{R}^3, \quad \mathbf{a} \cdot (\bar{\beta} \mathbf{a}) \geq |S| \left(\frac{a_1^2}{\kappa + \frac{4\mu}{3}} + \frac{a_2^2 + a_3^2}{\mu} \right) \quad (43)$$

where $|S| > 0$ denotes the dimensionless area of the void region S surrounding the corrugation within the elementary cell $\tilde{\Omega}$.

In the limiting case of symmetric corrugations along the direction x_2 , we have $\bar{b}_1^{\text{II}} = \bar{b}_2^{\text{I}} = 0$ and the effective tensor $\bar{\beta}$ becomes diagonal.

4.1.1. Proof of the symmetry of $\bar{\beta}$

From (30), multiplying $\tilde{\mathbf{v}} \cdot \tilde{\sigma}(\tilde{\mathbf{q}}^{\text{I}}) = \mathbf{0}$ by $\tilde{\mathbf{q}}^{\text{II}}$ and integrating over a finite subset domain $\tilde{\Omega}^* = \{\tilde{\mathbf{x}} \in \tilde{\Omega}; |\tilde{x}_1| < \tilde{x}_1^*\}$ of $\tilde{\Omega}$ with $\tilde{x}_1^* \gg 1$ taken to be large enough, we get after integration by parts that

$$0 = - \int_{\tilde{\Omega}^*} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{II}}) \cdot \tilde{\tau}^{\text{I}} d\tilde{\Omega} + \int_{\partial\tilde{\Omega}^*} \tilde{\mathbf{q}}^{\text{II}} \cdot (\tilde{\tau}^{\text{I}} \mathbf{n}) d\tilde{\Gamma}. \quad (44)$$

Given the asymptotic behavior of $\tilde{\tau}^{\text{I}}$ and $\tilde{\mathbf{q}}^{\text{II}}$ in (30) and (34) at infinity, the periodic boundary conditions on $\tilde{\Gamma}_{\#}$ and the Dirichlet/Neumann boundary conditions on $\tilde{\Gamma}_D$ and $\tilde{\Gamma}_N$, only the contribution at $\tilde{x}_1 = \tilde{x}_1^*$ is left in the second integral in (44) and we get

$$\int_{\tilde{\Omega}^*} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{II}}) \cdot \tilde{\tau}^{\text{I}} d\tilde{\Omega} = \left(\bar{\mathbf{b}}^{\text{II}} + \frac{\tilde{x}_1^*}{\mu} \mathbf{e}_1 \right) \cdot \left(\mathbf{e}_1 \otimes \mathbf{e}_1 + \frac{3\kappa - 2\mu}{3\kappa + 4\mu} \mathbf{e}_2 \otimes \mathbf{e}_2 \right) \mathbf{e}_1 + o(1) \quad (45)$$

$$= \bar{b}_1^{\text{II}} + o(1), \quad (46)$$

where $o(1)$ are vanishing terms when $\tilde{x}_1^* \rightarrow +\infty$. It now suffices to pass to the limit when $\tilde{x}_1^* \rightarrow +\infty$ to find

$$\bar{b}_1^{\text{II}} = \int_{\tilde{\Omega}} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{II}}) \cdot \tilde{\tau}^{\text{I}} d\tilde{\Omega} = \int_{\tilde{\Omega}} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{II}}) \cdot \mathbf{C} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{I}}) d\tilde{\Omega}. \quad (47)$$

Repeating the same calculation by exchanging the roles of I and II, we get

$$\bar{b}_2^{\text{I}} = \int_{\tilde{\Omega}} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{I}}) \cdot \tilde{\tau}^{\text{II}} d\tilde{\Omega} = \int_{\tilde{\Omega}} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{I}}) \cdot \mathbf{C} \tilde{\epsilon}(\tilde{\mathbf{q}}^{\text{II}}) d\tilde{\Omega}. \quad (48)$$

which, given the symmetry of the Hooke's law $\mathbf{C}$, proves $\bar{b}_2^{\text{I}} = \bar{b}_1^{\text{II}}$, hence the symmetry of $\bar{\beta}$.

4.1.2. Proof of the lower bound on $\bar{\beta}$

Given a vector $\mathbf{a} = (a_1, a_2, a_3) \in \mathbb{R}^3$, we consider the following vector fields $(\mathbf{v}, \mathbf{w})$ defined over $\tilde{\Omega}$ in the elementary cell,

$$\begin{cases} \mathbf{v}(\tilde{\mathbf{x}}) = a_1 \tilde{\mathbf{q}}^{\text{I}}(\tilde{\mathbf{x}}) + a_2 \tilde{\mathbf{q}}^{\text{II}}(\tilde{\mathbf{x}}) + a_3 \tilde{\mathbf{q}}^{\text{III}}(\tilde{\mathbf{x}}) \mathbf{e}_3 & \text{in } \tilde{\Omega}, \\ \mathbf{w}(\tilde{\mathbf{x}}) = \left(\frac{a_1}{\kappa + \frac{4\mu}{3}} \mathbf{e}_1 + \frac{a_2}{\mu} \mathbf{e}_2 + \frac{a_3}{\mu} \mathbf{e}_3 \right) \tilde{x}_1 & \text{in } \tilde{\Omega}, \end{cases} \quad (49)$$

with $\tilde{\mathbf{q}}^i$ ($i = \text{I, II, III}$) defined in (30) and (31). The field $\mathbf{v}$ hence satisfies the boundary value problem

$$\begin{cases} \tilde{\mathbf{v}} \cdot (\mathbf{C} \tilde{\epsilon}(\mathbf{v})) = \mathbf{0} & \text{in } \tilde{\Omega}, \\ \mathbf{v} \text{ } \tilde{x}_2\text{-periodic, } \mathbf{C} \tilde{\epsilon}(\mathbf{v}) \mathbf{n} \text{ } \tilde{x}_2\text{-anti-periodic} & \text{on } \tilde{\Gamma}_{\#}, \\ \mathbf{v} = \mathbf{0} & \text{on } \tilde{\Gamma}_D, \quad \mathbf{C} \tilde{\epsilon}(\mathbf{v}) \mathbf{n} = \mathbf{0} & \text{on } \tilde{\Gamma}_N, \end{cases} \quad (50)$$

with the following behavior at infinity for $\mathbf{v}$,

$$\mathbf{v} \underset{\tilde{x}_1 \rightarrow +\infty}{\sim} a_1 \left(\frac{\tilde{x}_1}{\kappa + \frac{4\mu}{3}} \mathbf{e}_1 + \bar{\mathbf{b}}^{\text{I}} \right) + a_2 \left(\frac{\tilde{x}_1}{\mu} \mathbf{e}_2 + \bar{\mathbf{b}}^{\text{II}} \right) + a_3 \left(\frac{\tilde{x}_1}{\mu} \mathbf{e}_3 + \bar{\mathbf{b}}^{\text{III}} \right) \mathbf{e}_3, \quad (51)$$

and for the corresponding stress $C\tilde{\epsilon}(\nu)$,

$$\lim_{\tilde{x}_1 \rightarrow +\infty} C\tilde{\epsilon}(\nu) = a_1 \left(e_1 \otimes e_1 + \frac{3\kappa - 2\mu}{3\kappa + 4\mu} (e_2 \otimes e_2 + e_3 \otimes e_3) \right) + a_2 (e_1 \otimes e_2 + e_2 \otimes e_1) + a_3 (e_1 \otimes e_3 + e_3 \otimes e_1). \quad (52)$$

We start by multiplying the divergence equation in (50) by ν and by integrating by parts over Ω^* to obtain

$$0 = \int_{\Omega^*} \nu \cdot (\tilde{\nabla} \cdot (C\tilde{\epsilon}(\nu))) d\tilde{\Omega} = - \int_{\Omega^*} \tilde{\epsilon}(\nu) \cdot C\tilde{\epsilon}(\nu) d\tilde{\Omega} + \int_{\partial\Omega^*} \nu \cdot (C\tilde{\epsilon}(\nu) e_1) d\tilde{\Gamma}. \quad (53)$$

Applying the boundary conditions satisfied by ν on $\tilde{\Gamma}_D$, $\tilde{\Gamma}_N$ and $\tilde{\Gamma}_\#$ given in (50) and using the behavior of the stress tensor $C\tilde{\epsilon}(\nu)$ at infinity in (52), it gives

$$\int_{\Omega^*} \tilde{\epsilon}(\nu) \cdot C\tilde{\epsilon}(\nu) d\tilde{\Omega} = \int_{\tilde{x}_1 = \tilde{x}_1^*} \nu \cdot (a_1 e_1 + a_2 e_2 + a_3 e_3) d\tilde{\Gamma} + o(1). \quad (54)$$

It then remains to plug the behavior of ν at infinity in (51) to obtain

$$\int_{\Omega^*} \tilde{\epsilon}(\nu) \cdot C\tilde{\epsilon}(\nu) d\tilde{\Omega} = a \cdot \bar{\beta} a + \left(\frac{a_1^2}{\kappa + \frac{4\mu}{3}} + \frac{a_2^2 + a_3^2}{\mu} \right) \tilde{x}_1^* + o(1). \quad (55)$$

Introducing the decomposition $\nu = w + (\nu - w)$ with w defined in (49) and expanding the left-hand side of (54) yields

$$\int_{\Omega^*} \tilde{\epsilon}(\nu) \cdot C\tilde{\epsilon}(\nu) d\tilde{\Omega} = \int_{\Omega^*} (\tilde{\epsilon}(\nu - w) \cdot C\tilde{\epsilon}(\nu - w) - \tilde{\epsilon}(w) \cdot C\tilde{\epsilon}(w) + 2\tilde{\epsilon}(w) \cdot C\tilde{\epsilon}(\nu)) d\tilde{\Omega} \quad (56)$$

Multiplying the divergence equation in (50) by w and applying the same arguments that lead to (55), we find

$$\int_{\Omega^*} \tilde{\epsilon}(w) \cdot C\tilde{\epsilon}(w) d\tilde{\Omega} = \left(\frac{a_1^2}{\kappa + \frac{4\mu}{3}} + \frac{a_2^2 + a_3^2}{\mu} \right) \tilde{x}_1^* + o(1), \quad (57)$$

while a direct calculation (given the analytical expression of w in (49)) shows that

$$\int_{\Omega^*} \tilde{\epsilon}(w) \cdot C\tilde{\epsilon}(w) d\tilde{\Omega} = \left(\frac{a_1^2}{\kappa + \frac{4\mu}{3}} + \frac{a_2^2 + a_3^2}{\mu} \right) (\tilde{x}_1^* - |S|), \quad (58)$$

with $|S|$ being the dimensionless area of the void region surrounding the corrugation. Substituting (57) and (58) in (56) and taking the limit as $\tilde{x}_1^* \rightarrow +\infty$, we obtain

$$a \cdot \bar{\beta} a = \int_{\Omega} \tilde{\epsilon}(\nu - w) \cdot C\tilde{\epsilon}(\nu - w) d\Omega + |S| \left(\frac{a_1^2}{\kappa + \frac{4\mu}{3}} + \frac{a_2^2 + a_3^2}{\mu} \right). \quad (59)$$

The first term of the right-hand side being positive, we deduce the lower bound (43).

4.1.3. Proof for symmetric corrugations: $\bar{b}_2^I = \bar{b}_1^{II} = 0$

For corrugations that are symmetric with respect to x_2 (or $\tilde{x}_2$ if working in the unit cell), the component $\tilde{q}_2^I(\tilde{x})$ of the generalized displacement $\tilde{q}^I(\tilde{x})$ solving the elementary problem (30) becomes odd with respect to $\tilde{x}_2$ as the loading consists in pulling along direction e_1 . Given the asymptotic behavior at infinity of $\tilde{q}^I(\tilde{x})$ in (34), we then must have $\bar{b}_2^I = 0$. Having established that the homogenized tensor $\bar{\beta}$ is symmetric, this implies also that $\bar{b}_1^{II} = 0$.

4.2. Weak formulation and uniqueness property of the homogenized problem

The homogenized macroscopic problem admits a weak formulation, making it suitable for finite element implementation. Moreover, the effective Robin-type boundary condition ensures the uniqueness of the solution, in the same way as the direct problem under Dirichlet boundary conditions. We state the following proposition:

Proposition. Let $C = \{\bar{u} : \bar{\Omega} \mapsto \mathbb{R}^3; \bar{u} = u^D \text{ on } \partial\bar{\Omega}_D\}$ and $C_0 = \{\hat{u} : \bar{\Omega} \mapsto \mathbb{R}^3; \hat{u} = 0 \text{ on } \partial\bar{\Omega}_D\}$ be the displacement solution space and weighting space for the homogenized problem, respectively. Assuming that the homogenized surface tensor $\bar{\beta}$ is invertible, the solution $u \in C$ is unique and satisfies the weak formulation

$$\int_{\bar{\Omega}} \epsilon(\bar{u}) \cdot C\epsilon(\hat{u}) d\Omega + \int_{\bar{\Gamma}} (\bar{u} - U_D) \cdot (\ell\bar{\beta})^{-1} \hat{u} dS - \int_{\partial\bar{\Omega}_N} t_N \cdot \hat{u} dS = 0, \quad \forall \hat{u} \in C_0. \quad (60)$$

The bulk energy of the corrugated region in the direct problem is thus replaced in the homogenized problem by a surface energy (on the boundary) that will combine with the classical bulk elastic energy to form the global stiffness matrix of the problem.

The weak formulation (60) is derived following the standard procedure of calculus of variation. Given a test function $\hat{u} \in C_0$, the equilibrium Eq. (1) is multiplied by $\hat{u}$ and integrated by parts. Making use of the constitutive relation in (2), together with the boundary conditions (4) and the effective boundary condition (5) (for which the positive definiteness of $\bar{\beta}$, established in Section 4.1, guarantees its invertibility), directly yields the weak form (60).

To establish the uniqueness of the homogenized problem, we consider two solutions $(\bar{u}^1, \bar{u}^2)$ of (60) and show that their difference vanishes identically in $\bar{\Omega}$. Subtracting the two weak formulations and taking $\hat{u} = \bar{u}^1 - \bar{u}^2$, we obtain

$$\int_{\bar{\Omega}} \epsilon(\bar{u}^1 - \bar{u}^2) : C \epsilon(\bar{u}^1 - \bar{u}^2) d\Omega + \int_{\bar{\Gamma}} (\bar{u}^1 - \bar{u}^2) \cdot (\ell \bar{\beta})^{-1} (\bar{u}^1 - \bar{u}^2) dS = 0, \quad (61)$$

which implies that both non-negative integrals must vanish. Since the integrand of the first term is positive definite, we deduce that $\epsilon(\bar{u}^1 - \bar{u}^2) = \mathbf{0}$ in $\bar{\Omega}$, so that $\bar{u}^1 - \bar{u}^2$ is a rigid body motion. Furthermore, knowing that $\bar{\beta}$ is positive definite (see (43)), we also have $\bar{u}^1 - \bar{u}^2 = \mathbf{0}$ on $\bar{\Gamma}$, which eliminates any rigid body motion. Hence, $\bar{u}^1 = \bar{u}^2$ in $\bar{\Omega}$, proving uniqueness.

5. Results

In this section, we assess the proposed effective model for the corrugated surface by comparing its predictions with direct full-field simulations. The 1st order homogenized problem is solved by implementing the weak formulation (60) in FEniCS (<https://fenicsproject.org>). For the sake of simplicity, we use as material parameters a normalized Young's modulus $E = 1$ and a Poisson's ratio $\nu = 0.3$, which imply $\mu = 0.385$ and $\kappa = 0.833$. Two types of periodicity are considered defined in terms of the small parameter $\eta = 1/5$ and $\eta = 1/15$, corresponding to $N = 5$ and $N = 15$ corrugations, respectively. The details of the full-field and microscopic unit-cell geometries as well as the resulting values for the corresponding homogenized $\bar{\beta}$ tensors used in this section are discussed in Appendix B. The rigid substrate bonded to the corrugated surface is assumed to be fixed, with zero prescribed displacement ($U_D = \mathbf{0}$), both in the full-field and the 1st order homogenized models.

For completeness, we also note here that in the full-field problem, we use a mesh with about 12×10^3 and 970×10^3 d.o.f in two- and three-dimensions, respectively. In turn, for the effective problem, given that the corrugation is replaced by an effective boundary condition on a flat surface, we use a mesh with about 3×10^3 and 55×10^3 in two- and three dimensions, respectively. It is evident that the acceleration factor especially in three-dimensions is substantial (estimated by direct calculations to more than 10 in two-dimensions and more than 100 in three-dimensions). For our simulations, we have used isoparametric quadratic four-node and quadratic tetrahedral elements in two- and three-dimensions, respectively. In passing, we note that in the following examples, we apply positive tensile loads combined with shear loads. Within the small-strain linear elasticity framework considered here, tensile and compressive loads yield identical responses, differing only by their sign.

5.1. Symmetric corrugation with normal stress only

In Fig. 3, we consider two cases of symmetric microstructures with $N = 5$ and $N = 15$ corrugations leading to 70% (in-plane surface) void fraction. In particular, Fig. 3 shows contour plots of the displacement field in both the undeformed and deformed configurations, comparing the full-field solution, the 0th order model, and the 1st order model. In this first example, we apply only $\sigma_{11} = 1.0$ at $x_1 = 1$. We observe an excellent agreement between the full-field results and the 1st order model using the homogenized Robin-type boundary condition (5), even in the case of very few corrugations, where the separation of length scales is not well preserved. In contrast, the 0th order model underestimates the displacement far from the fixed boundary. This discrepancy arises because the latter treats the interface as fully clamped (see Eq. (26)), leading to an underestimation of the displacement solution far away from the interface. The deformed plots for the 1st order model reveal that replacing the corrugated surface with an effective Robin-type boundary condition – where the traction is a function of displacement – yields a non-zero displacement at the interface with the rigid substrate. This displacement at the bottom is a homogenized representation of the local behavior observed near the interface in the full-field model. As the number of corrugations increases (while maintaining the same corrugation shape), i.e., as the small parameter ℓ decreases, the Dirichlet-type boundary conditions become increasingly dominant, and the 1st order model approaches the 0th order model. It is worth noting that the 0th order model yields the same result regardless of the size, number, or shape of the corrugations.

Fig. 4 shows line plots of displacement and stress magnitudes for the full-field solution, the 0th order model, and the 1st order model. In the full-field case, two locations within the corrugated region are considered to capture the geometric effects: one passing through a corrugation (AA') and the other through a void (BB'), as shown in Fig. 3. A normal traction load is applied at the top boundary ($x_1 = 1$), while the bottom boundary ($x_1 = 0$) is fixed. Near the corrugated interface, the stress magnitude along AA' in the full-field solution increases sharply due to the reduced effective material volume near the voids, while along BB', it drops significantly because of the traction-free void boundaries. In this region, the homogenized models – both the 0th and 1st order – fail to provide an accurate prediction. This is, however, expected, since the homogenized models serve to provide an average response in a zone where the full-field solution exhibits strong local variations, including stress singularities induced by the microstructure. Farther from the corrugated interface, the stress fields from both the 0th order and 1st order models converge towards the full-field solution, as they are constrained to satisfy the applied force at $x_1 = 1$. However, the estimate of the displacement field differs in the two models. The 1st order model shows excellent agreement with the full-field solution away from the interface, while the 0th order model significantly under-predicts the displacement. Conversely, if a prescribed displacement boundary condition were applied at $x_1 = 1$, the displacement would be well approximated by both the 0th order and 1st order models near the top boundary, whereas the stress would not.

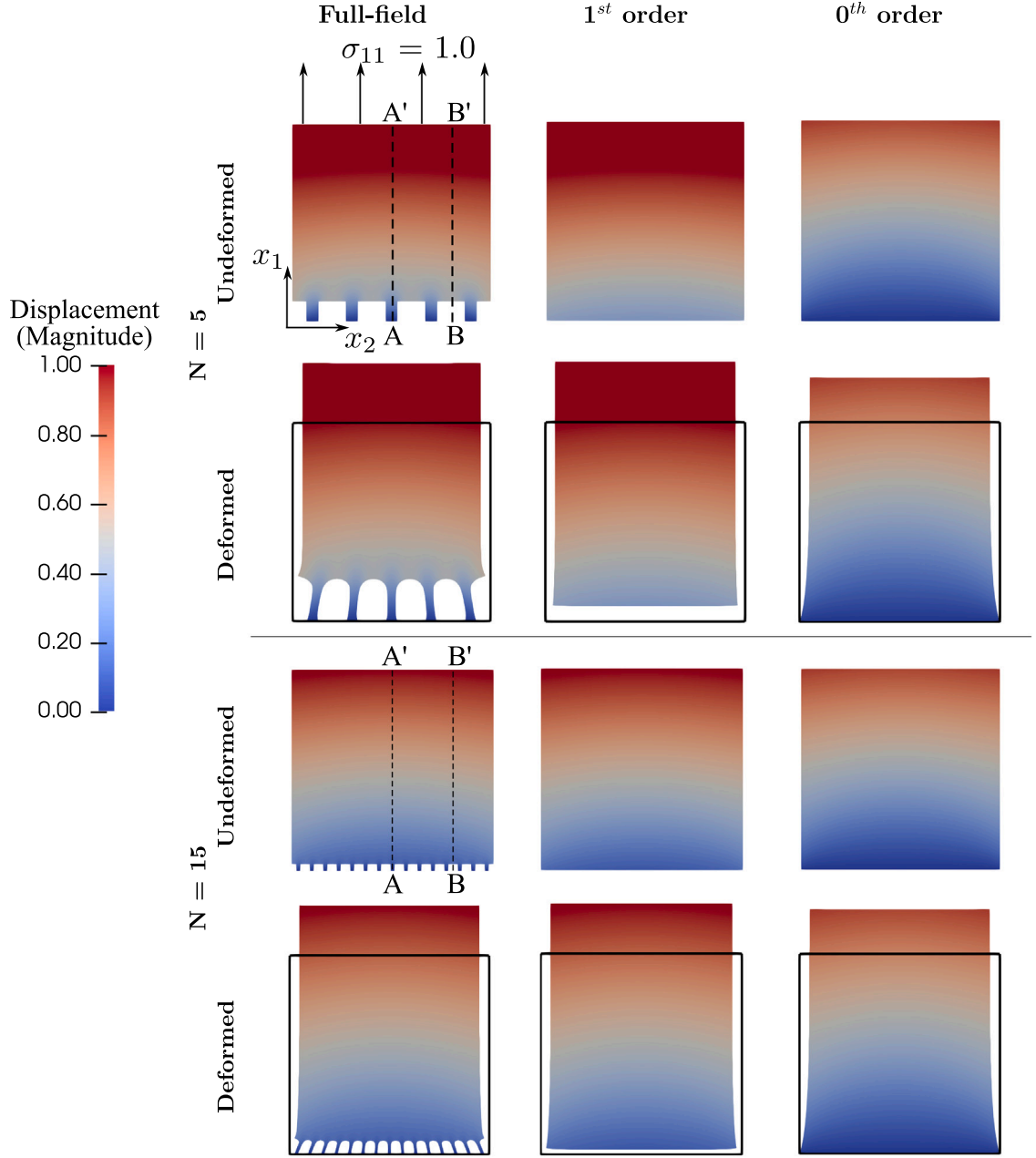


Fig. 3. Displacement fields obtained from full-field, effective and 0th order solution for the clamped corrugated boundary at $x_1 = 0$ and the normal traction at $x_1 = 1$ is $\sigma_{11} = 1.0$. Contour plots for displacement field are shown on the undeformed and deformed configuration for $N = 5$ and $N = 15$ corrugations.

Fig. 5 presents a convergence analysis comparing the full-field solution with the homogenized models (0th order and 1st order) by plotting the relative error as a function of the small parameter $\eta = \ell/L$. The test case involves an asymmetric corrugation subjected to a combined normal and shear loading ($\sigma_{11} = 1.0$, $\sigma_{12} = 0.5$) applied at the top boundary ($x_1 = 1$). The domain length is fixed at $L = 1$, while the periodicity ℓ is progressively reduced, corresponding to smaller values of η . The error is computed using the normalized L^2 norm

$$\text{Error} = \frac{\|\langle \mathbf{u}^{\text{FF}} \rangle - \langle \bar{\mathbf{u}} \rangle\|}{\|\langle \mathbf{u}^{\text{FF}} \rangle\|}, \quad (62)$$

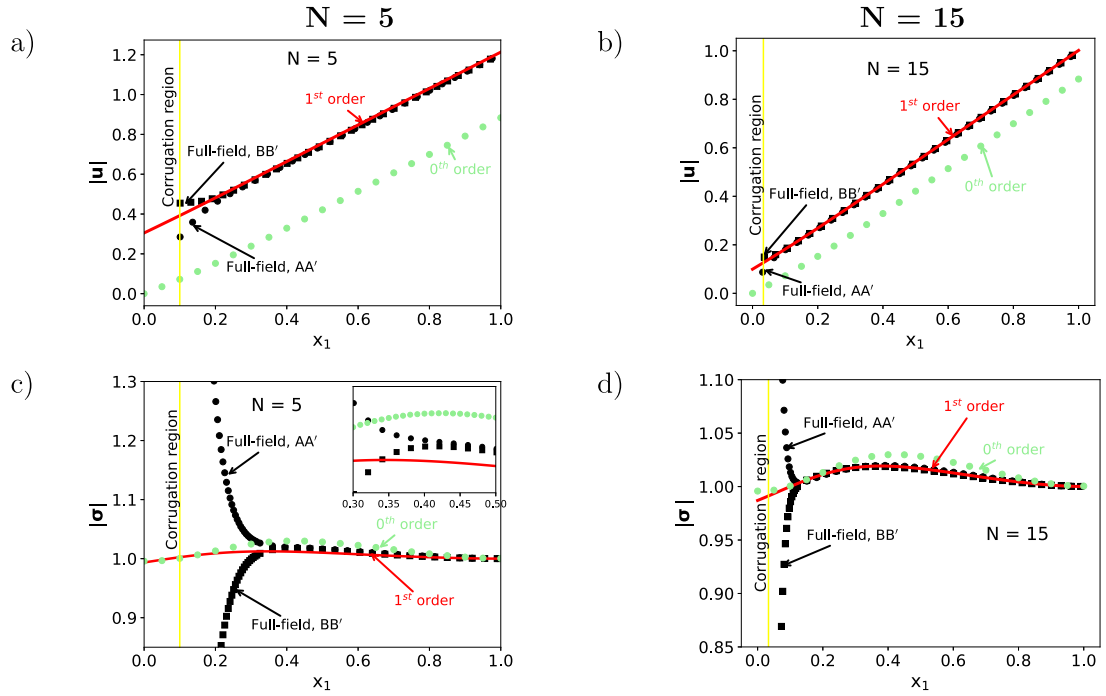


Fig. 4. Displacement and stress magnitude variation along the lines AA' and BB' in x_3 direction as shown in Fig. 3. AA' is in the corrugation region while BB' is in the void region. (a, b) displacement and (c, d) stress are obtained from full-field, effective and dominant order solution for $N = 5$ and $N = 15$ corrugations.

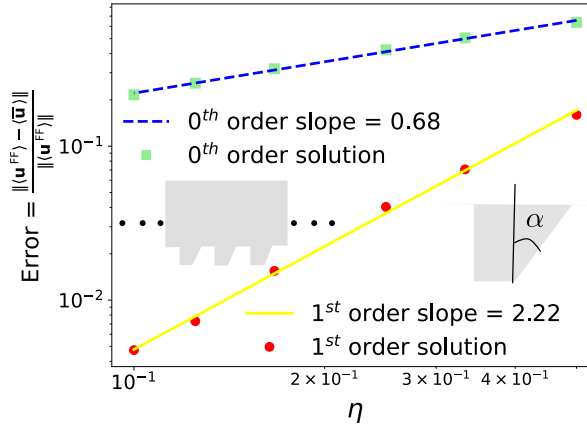


Fig. 5. Error convergence plot for 0th order and 1st order relative to the full-field result for an unsymmetrical corrugation with $\alpha = 20^\circ$. Combined normal and shear load as $\sigma_{11} = 1.0$ and $\sigma_{12} = 0.5$ was considered at $x_1 = 1$.

where $\langle u^{\text{FF}} \rangle$ denotes the average full-field displacement at the top boundary, and $\langle \bar{u} \rangle$ is the corresponding average displacement at the top boundary predicted by either the 0th order or 1st order model, depending on the case considered. As predicted by the asymptotic analysis (see (41)), the first-order model exhibits second-order convergence, while the 0th order model achieves only 1st order accuracy. Moreover, it is interesting to observe that even at larger values of η , where the separation of length scales starts losing its range of validity, the error remains fairly small thus allowing to use those solutions to estimate sufficiently well the effect of the corrugation upon the overall response of the structure.

5.2. Asymmetric corrugation with combined (normal and shear) stress at the top boundary

In the previous results, we observed that the far-field behavior is accurately captured by the 1st order homogenized model, whereas the 0th order model fails to provide a reliable prediction. However, in the vicinity of the corrugations, even the 1st order

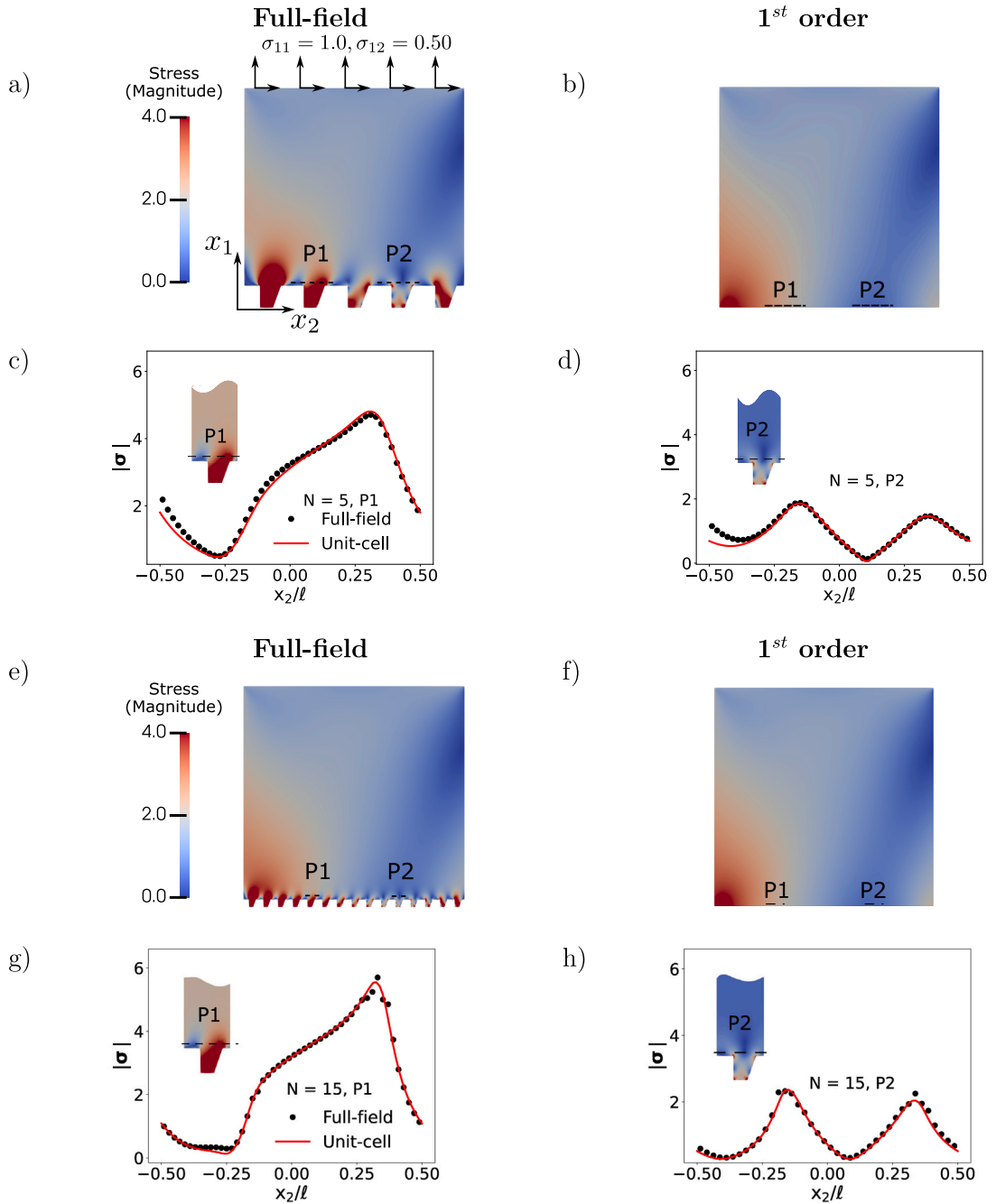


Fig. 6. Asymmetric corrugations. Comparison between the stress amplitude for the full-field and the reconstructed field near the corrugation region. The reconstructed stress field is obtained from post-processing analysis using (64). The traction at $x_1 = 1$ is $\sigma_{11} = 1.0$, $\sigma_{12} = 0.5$. Stress magnitude contour plots for (a), (b) $N=5$ corrugations and (e), (g) $N=15$ corrugations. Stress magnitude along P1 and P2 for (c), (d) $N=5$ corrugation, (g), (h) $N=15$ corrugations.

model yields only an “average” representation of the solution, within a region characterized by rapidly varying evanescent fields and stress concentrations at the corrugation tips. From a mechanical engineering standpoint, accessing detailed information in this critical region – without resorting to a full-field computation – would be highly beneficial for structural design. To this end, we propose a post-processing strategy that exploits the homogenization framework to reconstruct the local fields near the corrugated

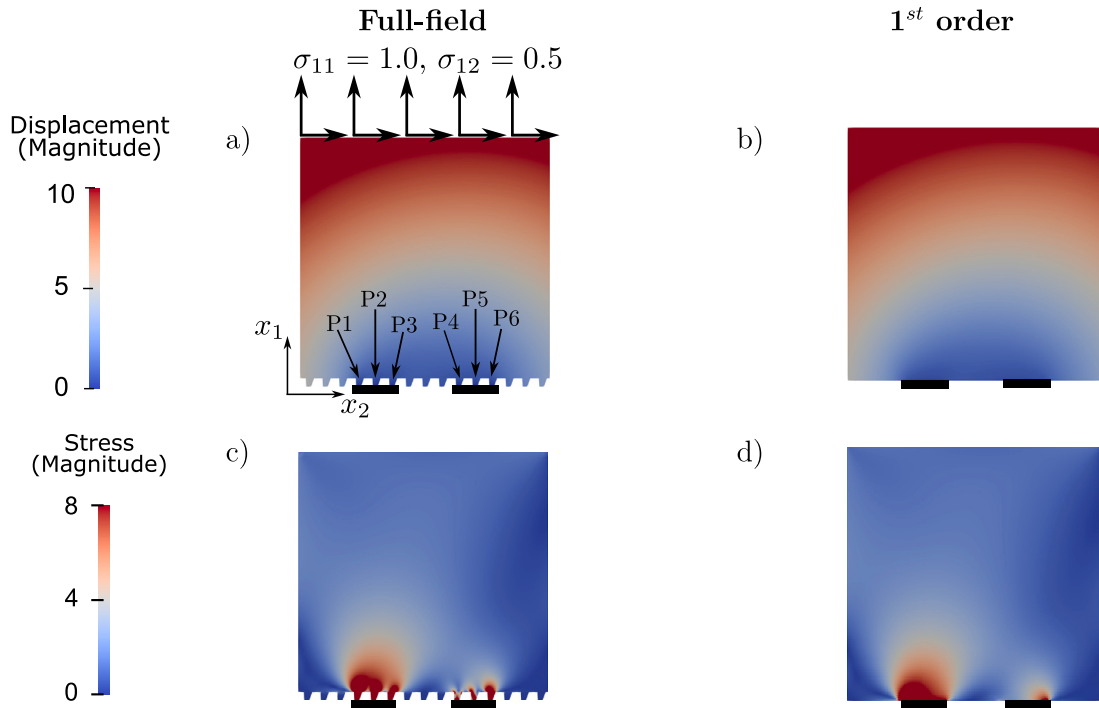


Fig. 7. Partially-applied Dirichlet boundary condition on the corrugated boundary i.e. only a fraction of the corrugations are subjected to Dirichlet boundary condition (shown by black patch) while others are stress-free. (a, b) Displacement magnitude contour plot for full-field and effective problem. (c, d) Stress magnitude contour plot for full-field and effective problem.

interface, based on the decomposition of the first-order generalized displacement field (35):

$$\tilde{u}(\tilde{x}, \mathbf{x}') = \bar{\sigma}_{11}(0, \mathbf{x}') \tilde{q}^I(\tilde{x}) + \bar{\sigma}_{12}(0, \mathbf{x}') \tilde{q}^{II}(\tilde{x}) e_2 + \bar{\sigma}_{13}(0, \mathbf{x}') \tilde{q}^{III}(\tilde{x}) e_3, \quad (63)$$

where $\tilde{u}(\tilde{x}, \mathbf{x}')$ is the local micro-displacement field computed at a point $\mathbf{x} = (0, \mathbf{x}')$ of the effective boundary $\bar{\Gamma}$ in the macroscopic domain $\bar{\Omega}$ and at a point $\tilde{x}$ in the rescaled unit cell. The functions $\tilde{q}^i(\tilde{x})$ ($i = I, II, III$) are the numerically precomputed solutions for the elementary cell problems (30) and (31) (already known from the computation of the effective coefficients). Their combination uses the macroscopic stress components $\bar{\sigma}_{1i}(0, \mathbf{x}')$ ($i = 1, 2, 3$), obtained from the 1st order homogenized model on a coarse mesh. Owing to the linearity of the constitutive relations, the corresponding local micro-stress field can then be computed via the constitutive relation (27):

$$\tilde{\sigma}(\tilde{x}, \mathbf{x}') = \bar{\sigma}_{11}(0, \mathbf{x}') C \tilde{\epsilon}(\tilde{q}^I)(\tilde{x}) + \bar{\sigma}_{12}(0, \mathbf{x}') C \tilde{\epsilon}(\tilde{q}^{II})(\tilde{x}) + \bar{\sigma}_{13}(0, \mathbf{x}') C \tilde{\epsilon}(\tilde{q}^{III} e_3)(\tilde{x}). \quad (64)$$

To assess the accuracy of this post-processing strategy, we present stress field contour plots comparing the full-field results with the reconstructed local fields, together with cross-line plots of stress magnitude along selected paths, in Fig. 6. The problem is solved for two cases: $N = 5$ corrugations (a-d) and $N = 15$ corrugations (e-h). In both cases, the fields are plotted at the same macroscopic location just above the corrugated region. The paths used for comparison are marked as P1 and P2 in the contour plots. For both configurations ($N = 5$ and 15), the reconstructed local fields show excellent agreement with the full-field solution. This reveals the capability to accurately recover local information without resorting to fine-scale full-field simulations of the entire structure and thus represents a critical advantage of the proposed method, particularly in reducing computational costs.

5.3. Non-uniformly applied Dirichlet conditions

In all previous cases, we assumed that the corrugated surface was entirely bonded to the rigid substrate, resulting in a uniform Dirichlet boundary condition at $x_1 = 0$. However, in certain problems of practical interest, it may be relevant to consider cases where only a subset of the corrugations is in contact with the substrate, while the rest remain traction-free. In such cases, a good approximation can be achieved by applying the first-order effective boundary condition (5) locally, restricted to the contact region. The remaining corrugations, which are not in contact with the substrate, are replaced by a flat boundary with traction-free boundary condition.

This is expected to correspond to the 0th order homogenized boundary condition for traction-free corrugations (see Marigo and Maurel, 2016 for a related acoustic case, which is equivalent to the anti-plane elasticity setting). Fig. 7 presents a schematic view of

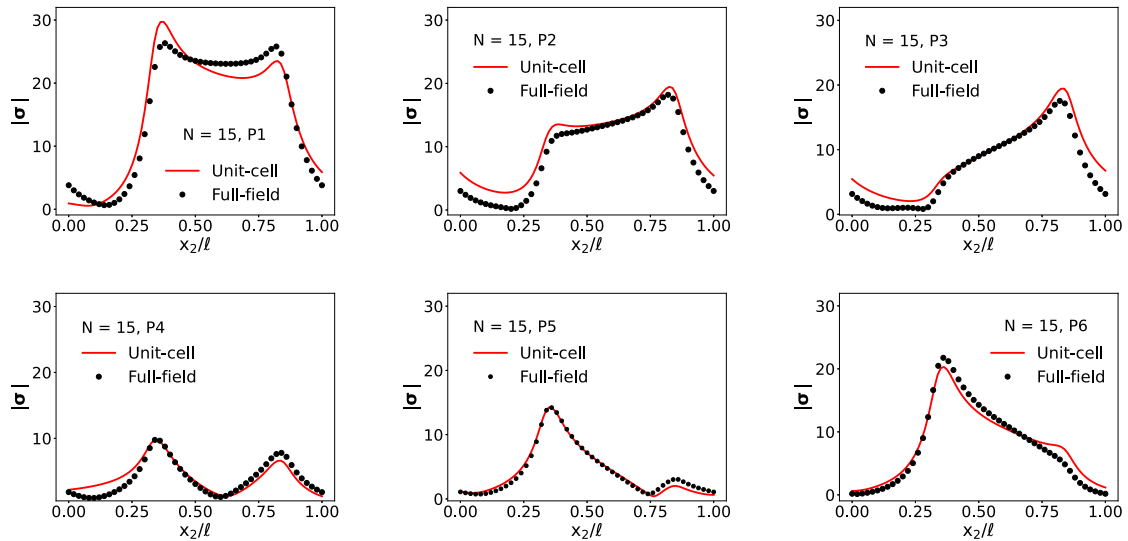


Fig. 8. Comparison of the local stress field obtained from full-field simulation and the reconstructed local field from the first order homogenized results in the case of non-uniformly applied Dirichlet boundary conditions.

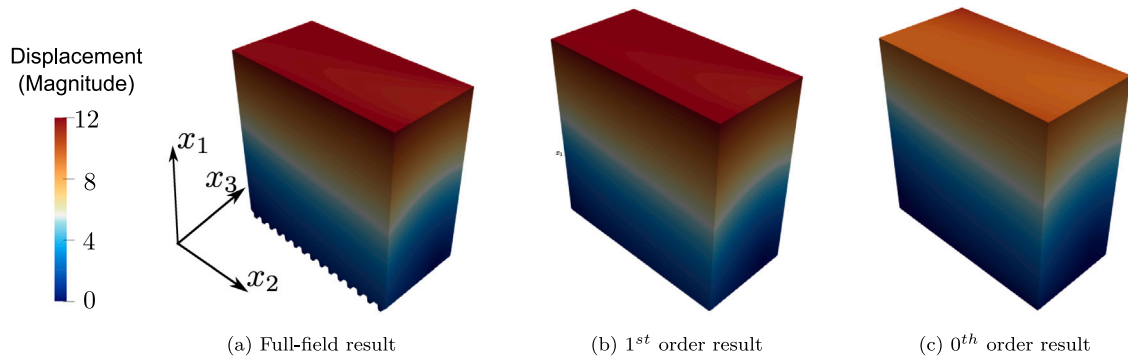


Fig. 9. 3D implementation of the proposed model where the top surface is subjected to traction $\sigma_{11} = 1.0$, $\sigma_{12} = 0.5$, $\sigma_{13} = 0.5$. Contour plot for displacement field under mixed in-plane and out-of-plane loading is plotted for (a) full-field solution, (b) 1st order solution and (c) 0th order solution.

the setup, along with displacement and stress contour plots for a corrugated surface subject to a locally-applied Dirichlet condition. The black patches indicate the region where the surface is clamped, while the remaining portion is traction-free. The displacement fields predicted by the homogenized model show very good agreement with the full-field results. This may be attributed to the observation that the 1st order model works sufficiently well even for smaller number of corrugations (e.g. larger values of η), where there is no clear separation of length scales (see discussion in the context of Fig. 5). In addition, the stress contours confirm that the effective model accurately captures the localized stress concentrations near the clamped region (see Figs. 7c and 7d).

Fig. 8 compares the local fields obtained from full-field simulations with those reconstructed using the homogenized model in the 2D case. In the vicinity of corrugations P1, P2, and P3, where stress gradients are pronounced, the first-order homogenized model yields very good estimates of the local response. In regions near corrugations P4, P5, and P6, where the gradients are milder, the agreement is excellent. For problems involving extremely steep gradients at the macroscopic scale, particularly near corners where singularities arise, accurate reconstruction of the local fields may require incorporating higher-order terms in the asymptotic expansions, although this would inevitably increase both the computational cost and the algebraic complexity.

5.4. Three-dimensional simulation under mixed loading

For completeness and after thoroughly investigating the method for in-plane problems, we present next a three-dimensional simulation of an elastic body with corrugations invariant along the x_3 direction. In this case, we employ the full form of the first-order effective boundary condition (5), which includes the effective coefficient $\bar{b}^{\text{III}}$ accounting for out-of-plane effects. To construct

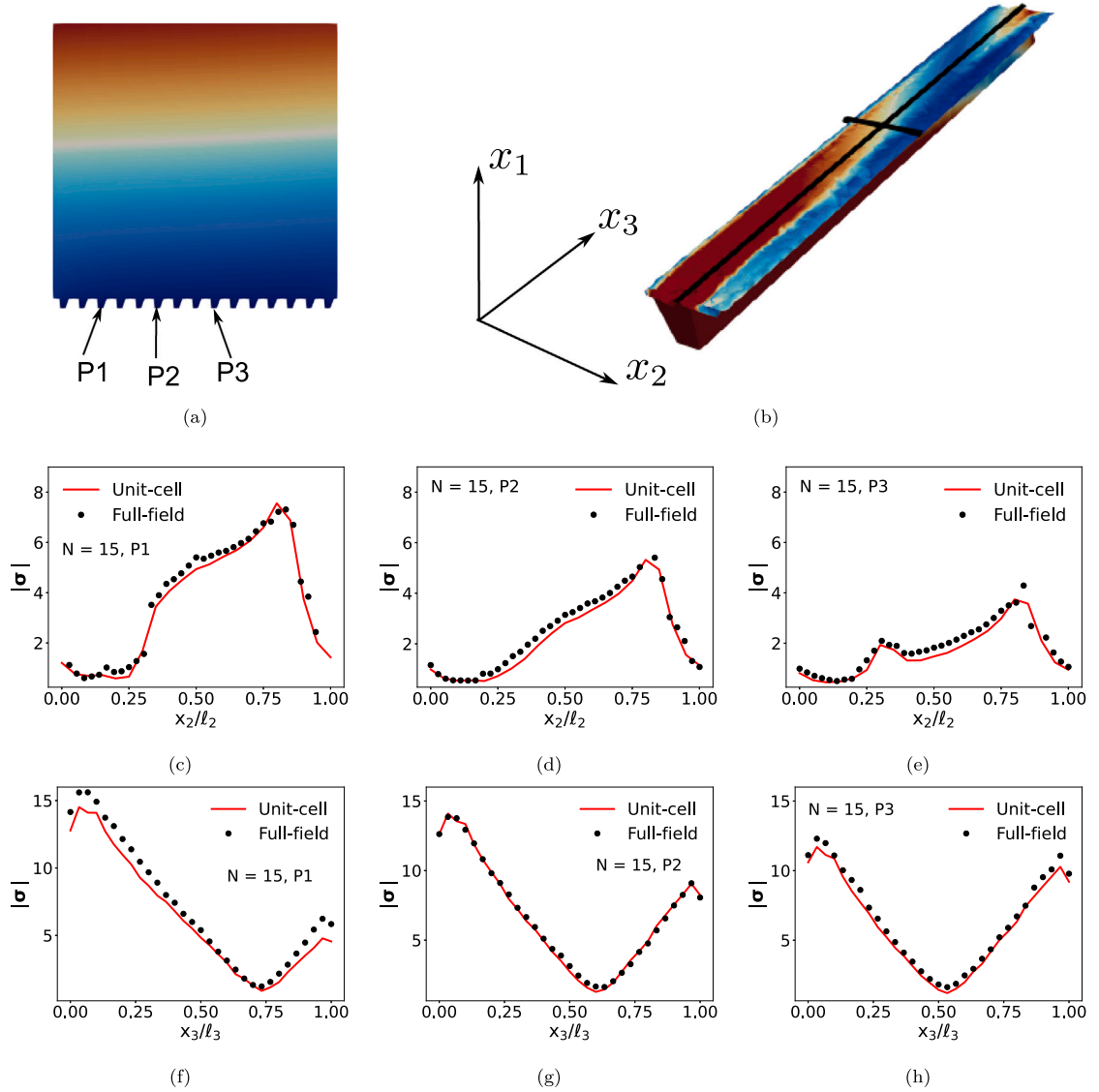


Fig. 10. Comparison of the local stress field obtained from full-field simulation and the reconstructed local field from the first order homogenization estimates for the 3D problem under mixed loading.

the 3D geometry, we extrude along the x_3 direction the structure previously used. One could consider a more complex periodic pattern in both (x_2 and x_3) directions or even a random one (with a larger unit-cell), however such an endeavor has not been pursued here and will be considered in future works with more applied problems at hand. At the top surface ($x_1 = 1$), we apply the normalized stress components $\sigma_{11} = 1.0$, $\sigma_{12} = 0.5$, and $\sigma_{13} = 0.5$. The resulting displacement magnitude is shown in Fig. 9. Once again, the 1st order model exhibits excellent agreement with the full-field solution, while the 0th order model fails to accurately capture the displacement at the top surface.

For additional quantitative validation of the proposed model in a 3D setting, Fig. 10 presents a comparison between the stress magnitude obtained from full-field simulations and the 1st order homogenized estimates. The comparison is shown along the centerlines in the x_2 and x_3 directions for the 3rd, 6th, and 9th corrugations (see Fig. 10(a)) counting from the left (black lines in Fig. 10(b)). The line plots in the x_2 10(c)–10(e) and x_3 10(f)–10(h) directions demonstrate an excellent agreement between the 1st order homogenized predictions and the full-field results.

6. Conclusion

In the present study, we have developed an effective interface model in linear elasticity for periodic corrugated boundaries bonded to a rigid substrate, in which the detailed corrugated geometry is replaced by a zero-thickness interface governed by homogenized Robin-type boundary conditions. The effective model is derived through an asymptotic analysis carried out up to first order, combining periodic homogenization techniques with matched asymptotic expansions to account for boundary-layer effects induced by the microstructure.

A key outcome of the analysis is the emergence of a symmetric, positive-definite second-order effective tensor that encapsulates the mechanical influence of surface heterogeneities by linking the interfacial displacement field to the corresponding traction. The resulting effective problem admits a variational formulation featuring a nonstandard surface energy term, in addition to the classical bulk contribution, ensuring full compatibility with standard finite element discretizations. In particular, the analysis leads to a linear contact or adhesive law (depending on the sign of the macroscopically applied load) that depends directly on the geometrical characteristics of the surface corrugations.

The model is validated against 2D and 3D full-field simulations in several representative configurations, showing excellent agreement in both displacement and stress fields, with second-order convergence consistent with theoretical predictions. Furthermore, the asymptotic framework enables the reconstruction of near-field variations in the corrugated layer through an efficient post-processing strategy, thereby significantly reducing computational cost.

These results provide a robust and versatile framework for modeling corrugated elastic surfaces and interfaces, and open perspectives for further developments, including extensions to more complex geometries (Yastrebov et al., 2014), constitutive behaviors, and adhesion laws. Of particular interest would be the study of rough surfaces (Yastrebov et al., 2015; Vakis et al., 2018) or material interfaces (Fleck and Willis, 2016) with random features as well as the rigorous scale-up of realistic adhesive or contact laws via interfacial homogenization (Sánchez-Palencia and Suquet, 1983). The first study could be readily carried out in the linear elastic context by replacing the present periodic corrugations with random surface roughness, requiring, however, proper ergodicity assumptions and representative “surface” elements for a proper homogenized solution. The second study of extracting realistic nonlinear contact or adhesive laws is much less straightforward and most probably requires an elaborate extension of the proposed approach in nonlinear materials and perhaps finite strains. Such an endeavor is underway but as in bulk homogenization, it is expected to require a nontrivial amount of time and effort.

CRedit authorship contribution statement

Vivek Singh: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Kim Pham:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Arthur Geromel Fischer:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Kostas Danas:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Vivek Singh and partially Kim Pham and Kostas Danas report financial support was provided by Manufacture Française des Pneumatiques Michelin SAS. Arthur Geromel-Fischer is an employee of the aforementioned company.

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Appendix A. Numerical computation of the effective tensor $\bar{\beta}$

In theory, the computation of the effective parameters $(\bar{b}^I, \bar{b}^{II}, \bar{b}^{III})$ in (7)–(8), which enter the second-order effective tensor $\bar{\beta}$, requires solving the elementary problems (30)–(31) on a *semi-infinite* periodic cell $\tilde{\Omega}$. In practice, these problems are solved over a truncated domain $\tilde{\Omega}^* = \{\tilde{x} \in \tilde{\Omega}; |\tilde{x}_1| < \tilde{x}_1^*\}$, where the upper limit $\tilde{x}_1^*$ is taken sufficiently large – typically around five times the corrugation height – to ensure convergence of the solution toward its asymptotic behavior. A weak formulation of (30)–(31) is then employed for numerical implementation.

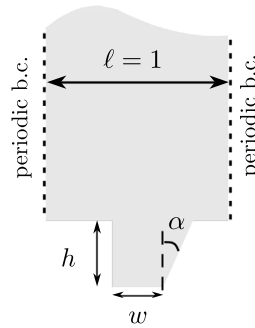
Let $\tilde{C}_0^I = \{\hat{q} : \tilde{\Omega} \mapsto \mathbb{R}^3; \hat{q} = \mathbf{0} \text{ on } \tilde{\Gamma}_D\}$ ($i=I,II$) and $\tilde{C}_0^{III} = \{\hat{q} : \tilde{\Omega} \mapsto \mathbb{R}; \hat{q} = 0 \text{ on } \tilde{\Gamma}_D\}$ be the weighting spaces associated with the in-plane and out-of-plane elementary problems, respectively. Multiplying the governing equations by admissible test functions, integrating by parts and applying the boundary/periodic conditions yields the following weak formulations for the in-plane problems

$$\int_{\tilde{\Omega}^*} \tilde{\epsilon}^{2D}(\tilde{q}_{\text{approx}}^i) \cdot C\tilde{\epsilon}^{2D}(\hat{q})d\Omega - \int_{-\frac{1}{2}}^{\frac{1}{2}} e_i \cdot \hat{q}(\tilde{x}_1^*, \tilde{x}_2)d\tilde{x}_2 = 0, \quad \forall \hat{q} \in \tilde{C}_0^i, \quad (\text{A.1})$$

Table B.1

Geometrical parameters used in the full-field and unit-cell geometry for the symmetric and asymmetric corrugations.

	N	$L = N\ell$	$\eta = \ell/L$	ℓ	h	w	α
Symmetric corrug.	5	5.0	1/5	1.0	0.5	0.3	0.0
	15	15.0	1/15	1.0	0.5	0.3	0.0
Asymmetric corrug.	5	5.0	1/5	1.0	0.5	0.3	20.0
	15	15.0	1/15	1.0	0.5	0.3	20.0

**Fig. B.11.** Schematic of the non-dimensional asymmetric unit-cell geometry. The symmetric corrugation can be obtained by setting the angle $\alpha = 0$.

with $(i, i) = (I, 1)$ or $(II, 2)$; and for the out-of-plane problem,

$$\int_{\tilde{\Omega}^*} \mu \tilde{\nabla}^{2D}(\tilde{q}_{\text{approx}}^{\text{III}}) \cdot \tilde{\nabla}^{2D}(\hat{q}) d\tilde{\Omega} - \int_{-\frac{1}{2}}^{\frac{1}{2}} \hat{q}(\tilde{x}_1^*, \tilde{x}_2) d\tilde{x}_2 = 0, \quad \forall \hat{q} \in \tilde{C}_0^{\text{III}}. \quad (\text{A.2})$$

Once the weak formulations have been solved – typically using the finite element method – the effective parameters can be computed by evaluating the following approximations at the truncation boundary $\tilde{x}_1 = \tilde{x}_1^*$:

$$\begin{cases} \bar{b}_{\text{approx}}^{\text{I}} = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\tilde{q}_{\text{approx}}^{\text{I}}(\tilde{x}_1^*, \tilde{x}_2) - \frac{\tilde{x}_1^*}{\kappa + \frac{4\mu}{3}} e_1 \right) d\tilde{x}_2, & \bar{b}_{\text{approx}}^{\text{II}} = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\tilde{q}_{\text{approx}}^{\text{II}}(\tilde{x}_1^*, \tilde{x}_2) - \frac{\tilde{x}_1^*}{\mu} e_2 \right) d\tilde{x}_2, \\ \bar{b}_{\text{approx}}^{\text{III}} = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\tilde{q}_{\text{approx}}^{\text{III}}(\tilde{x}_1^*, \tilde{x}_2) - \frac{\tilde{x}_1^*}{\mu} \right) d\tilde{x}_2. \end{cases} \quad (\text{A.3})$$

Appendix B. Geometric details and $\bar{\beta}$ tensor for symmetric and asymmetric corrugation

In this section, we provide some details on the geometrical parameters used in our simulations. Given the theoretical character of the analysis, all dimensions may be considered inconsequential and thus are non-dimensionalized. With reference to the dimensions defined in Fig. 1 and Fig. B.11, we define first the dimension of the microscopic problem by fixing the width of the unit cell to be $\ell = 1$. We then find that the coefficients of the $\bar{\beta}$ converge for a height of the microscopic unit cell that is equal to 5ℓ . Then, the width of the structure in full-field problem is simply $L = \ell N$, with N denoting the total number of corrugations. In addition, we consider the height of the structure in the full-field problem to be equal to L leading to a square overall domain. In turn, the corrugation geometry is totally defined by ℓ , its height h and width w in the symmetric case, while in the asymmetric case a non-zero angle α is also introduced. In Table B.1, we provide specific choices for all these parameters used in our calculations in the present work. For the three-dimensional case, we simply extrude the two-dimensional geometry along direction e_3 (see Fig. 1) using the same dimension L .

The effective tensors $\bar{\beta}$ are estimated for the aforementioned cases and for both N to take the values

$$\begin{cases} \text{symm. :} & \bar{\beta} = 1.54 e_1 \otimes e_1 + 10.88 e_2 \otimes e_2 + 3.72 e_3 \otimes e_3 \\ \text{asymm. :} & \bar{\beta} = 1.14 e_1 \otimes e_1 + 6.53 e_2 \otimes e_2 + 2.55 e_3 \otimes e_3 - 0.80 (e_1 \otimes e_2 + e_2 \otimes e_1). \end{cases}$$

This result implies that the microstructural tensor $\bar{\beta}$ converges even for $\eta = 1/5$ (i.e. $N = 5$), which corresponds to a fairly low number of corrugations and thus separation of length scales is not strictly preserved since $\ell = 0.2L$ which is of order L in this case.

Data availability

No data was used for the research described in the article.

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